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4.4 SHORTEST PATHS

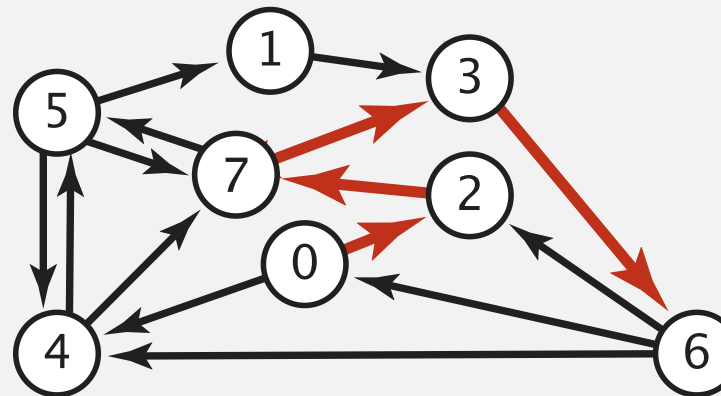
- ▶ *APIs*
- ▶ *shortest-paths properties*
- ▶ *Dijkstra's algorithm*
- ▶ *edge-weighted DAGs*
- ▶ *negative weights*

Shortest paths in an edge-weighted digraph

Given an edge-weighted digraph, find the shortest path from s to t .

edge-weighted digraph

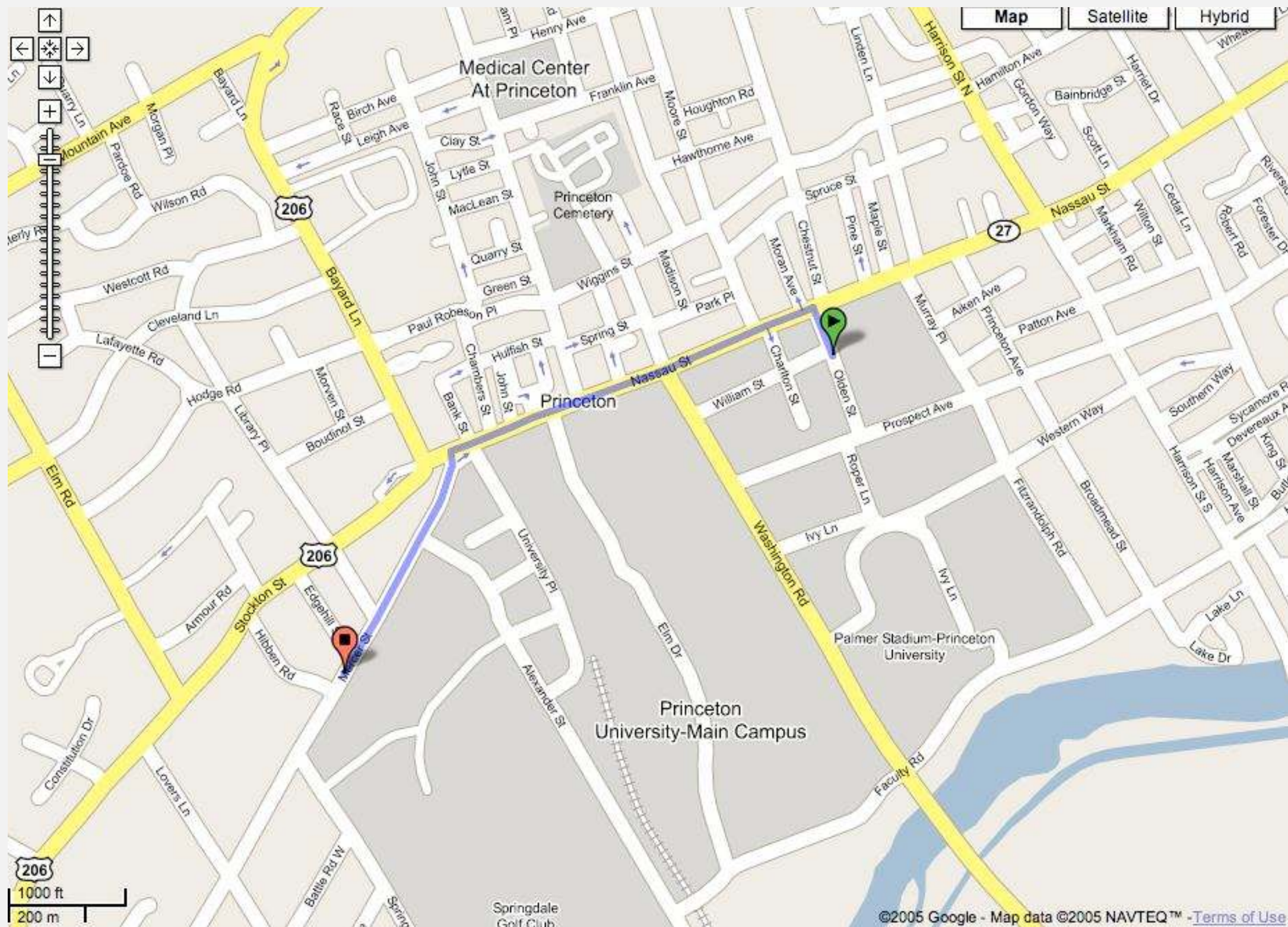
4→5	0.35
5→4	0.35
4→7	0.37
5→7	0.28
7→5	0.28
5→1	0.32
0→4	0.38
0→2	0.26
7→3	0.39
1→3	0.29
2→7	0.34
6→2	0.40
3→6	0.52
6→0	0.58
6→4	0.93



shortest path from 0 to 6

0→2	0.26
2→7	0.34
7→3	0.39
3→6	0.52

Google maps



Shortest path applications

- PERT/CPM.
- Map routing.
- **Seam carving**.
- Texture mapping.
- Robot navigation.
- Typesetting in TeX.
- Urban traffic planning.
- Optimal pipelining of VLSI chip.
- Telemarketer operator scheduling.
- Routing of telecommunications messages.
- Network routing protocols (OSPF, BGP, RIP).
- Exploiting **arbitrage** opportunities in currency exchange.
- Optimal truck routing through given traffic congestion pattern.



http://en.wikipedia.org/wiki/Seam_carving



Reference: Network Flows: Theory, Algorithms, and Applications, R. K. Ahuja, T. L. Magnanti, and J. B. Orlin, Prentice Hall, 1993.

Shortest path variants

Which vertices?

- **Single source:** from one vertex s to every other vertex.
- Single sink: from every vertex to one vertex t .
- Source-sink: from one vertex s to another t .
- All pairs: between all pairs of vertices.

Restrictions on edge weights?

- Nonnegative weights.
- Euclidean weights.
- Arbitrary weights.

Cycles?

- No directed cycles.
- No "negative cycles."



which variant?

Simplifying assumption. Shortest paths from s to each vertex v exist.



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Weighted directed edge API

```
public class DirectedEdge
```

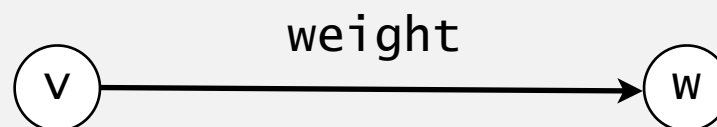
```
    DirectedEdge(int v, int w, double weight)    weighted edge  $v \rightarrow w$ 
```

```
    int from()                                    vertex  $v$ 
```

```
    int to()                                       vertex  $w$ 
```

```
    double weight()                              weight of this edge
```

```
    String toString()                            string representation
```



Idiom for processing an edge e : `int v = e.from(), w = e.to();`

Weighted directed edge: implementation in Java

Similar to Edge for undirected graphs, but a bit simpler.

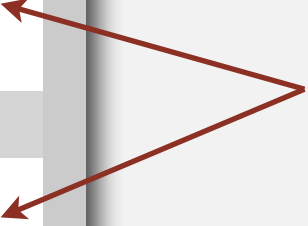
```
public class DirectedEdge
{
    private final int v, w;
    private final double weight;

    public DirectedEdge(int v, int w, double weight)
    {
        this.v = v;
        this.w = w;
        this.weight = weight;
    }

    public int from()
    { return v; }

    public int to()
    { return w; }

    public int weight()
    { return weight; }
}
```



from() and to() replace
either() and other()

Edge-weighted digraph API

```
public class EdgeWeightedDigraph
```

```
    EdgeWeightedDigraph(int V)    edge-weighted digraph with  $V$  vertices
```

```
    EdgeWeightedDigraph(In in)    edge-weighted digraph from input stream
```

```
    void addEdge(DirectedEdge e)    add weighted directed edge  $e$ 
```

```
    Iterable<DirectedEdge> adj(int v)    edges pointing from  $v$ 
```

```
    int V()    number of vertices
```

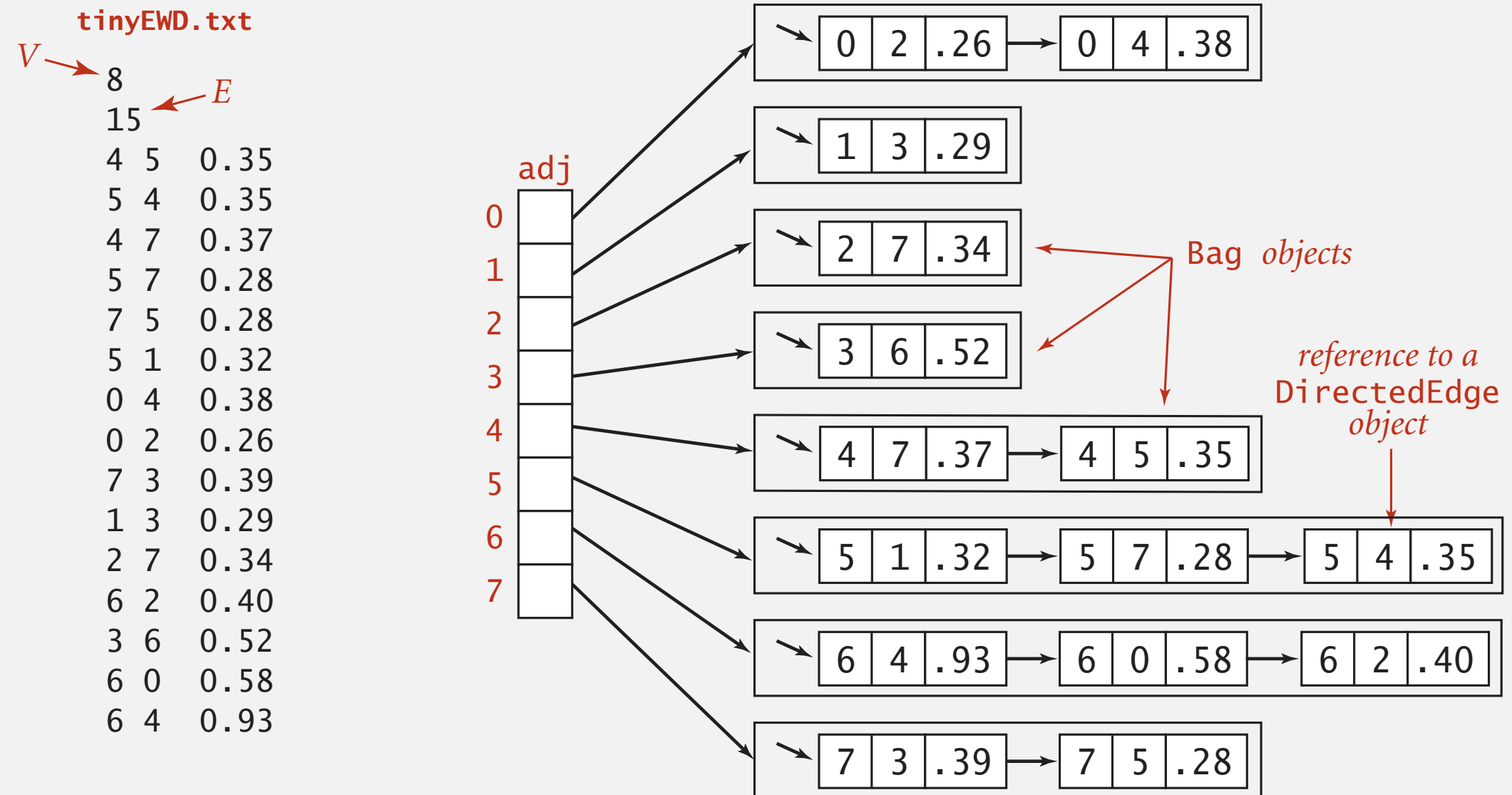
```
    int E()    number of edges
```

```
    Iterable<DirectedEdge> edges()    all edges
```

```
    String toString()    string representation
```

Conventions. Allow self-loops and parallel edges.

Edge-weighted digraph: adjacency-lists representation



Edge-weighted digraph: adjacency-lists implementation in Java

Same as EdgeWeightedGraph except replace Graph with Digraph.

```
public class EdgeWeightedDigraph
{
    private final int V;
    private final Bag<DirectedEdge>[] adj;

    public EdgeWeightedDigraph(int V)
    {
        this.V = V;
        adj = (Bag<DirectedEdge>[]) new Bag[V];
        for (int v = 0; v < V; v++)
            adj[v] = new Bag<DirectedEdge>();
    }

    public void addEdge(DirectedEdge e)
    {
        int v = e.from();
        adj[v].add(e);
    }

    public Iterable<DirectedEdge> adj(int v)
    { return adj[v]; }
}
```

← add edge $e = v \rightarrow w$ to
only v 's adjacency list

Single-source shortest paths API

Goal. Find the shortest path from s to every other vertex.

```
public class SP
```

```
    SP(EdgeWeightedDigraph G, int s) shortest paths from s in graph G
```

```
    double distTo(int v) length of shortest path from s to v
```

```
    Iterable <DirectedEdge> pathTo(int v) shortest path from s to v
```

```
    boolean hasPathTo(int v) is there a path from s to v?
```

```
SP sp = new SP(G, s);
for (int v = 0; v < G.V(); v++)
{
    StdOut.printf("%d to %d (%.2f): ", s, v, sp.distTo(v));
    for (DirectedEdge e : sp.pathTo(v))
        StdOut.print(e + " ");
    StdOut.println();
}
```


Single-source shortest paths API

Goal. Find the shortest path from s to every other vertex.

```
public class SP
```

```
    SP(EdgeWeightedDigraph G, int s)    shortest paths from s in graph G
```

```
    double distTo(int v)                length of shortest path from s to v
```

```
    Iterable <DirectedEdge> pathTo(int v)    shortest path from s to v
```

```
    boolean hasPathTo(int v)            is there a path from s to v?
```

```
% java SP tinyEWD.txt 0
0 to 0 (0.00):
0 to 1 (1.05): 0->4 0.38 4->5 0.35 5->1 0.32
0 to 2 (0.26): 0->2 0.26
0 to 3 (0.99): 0->2 0.26 2->7 0.34 7->3 0.39
0 to 4 (0.38): 0->4 0.38
0 to 5 (0.73): 0->4 0.38 4->5 0.35
0 to 6 (1.51): 0->2 0.26 2->7 0.34 7->3 0.39 3->6 0.52
0 to 7 (0.60): 0->2 0.26 2->7 0.34
```



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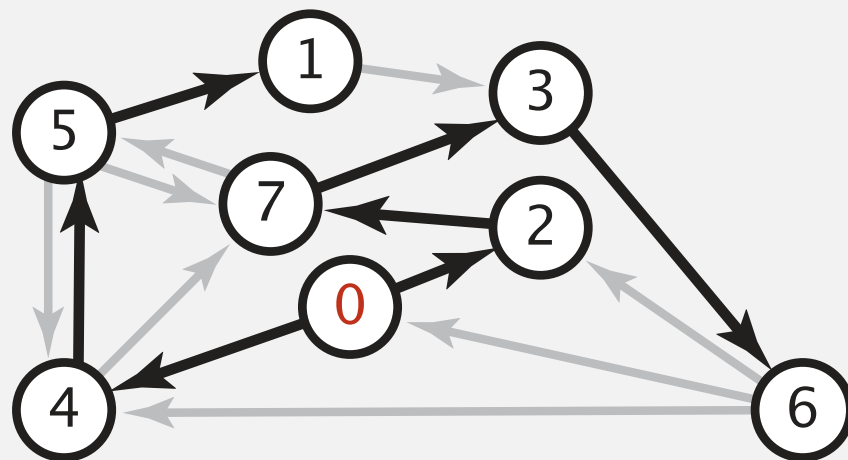
Data structures for single-source shortest paths

Goal. Find the shortest path from s to every other vertex.

Observation. A **shortest-paths tree** (SPT) solution exists. Why?

Consequence. Can represent the SPT with two vertex-indexed arrays:

- $\text{distTo}[v]$ is length of shortest path from s to v .
- $\text{edgeTo}[v]$ is last edge on shortest path from s to v .



shortest-paths tree from 0

	edgeTo[]	distTo[]
0	null	0
1	5->1 0.32	1.05
2	0->2 0.26	0.26
3	7->3 0.37	0.97
4	0->4 0.38	0.38
5	4->5 0.35	0.73
6	3->6 0.52	1.49
7	2->7 0.34	0.60

parent-link representation

Data structures for single-source shortest paths

Goal. Find the shortest path from s to every other vertex.

Observation. A **shortest-paths tree** (SPT) solution exists. Why?

Consequence. Can represent the SPT with two vertex-indexed arrays:

- `distTo[v]` is length of shortest path from s to v .
- `edgeTo[v]` is last edge on shortest path from s to v .

```
public double distTo(int v)
{   return distTo[v];   }

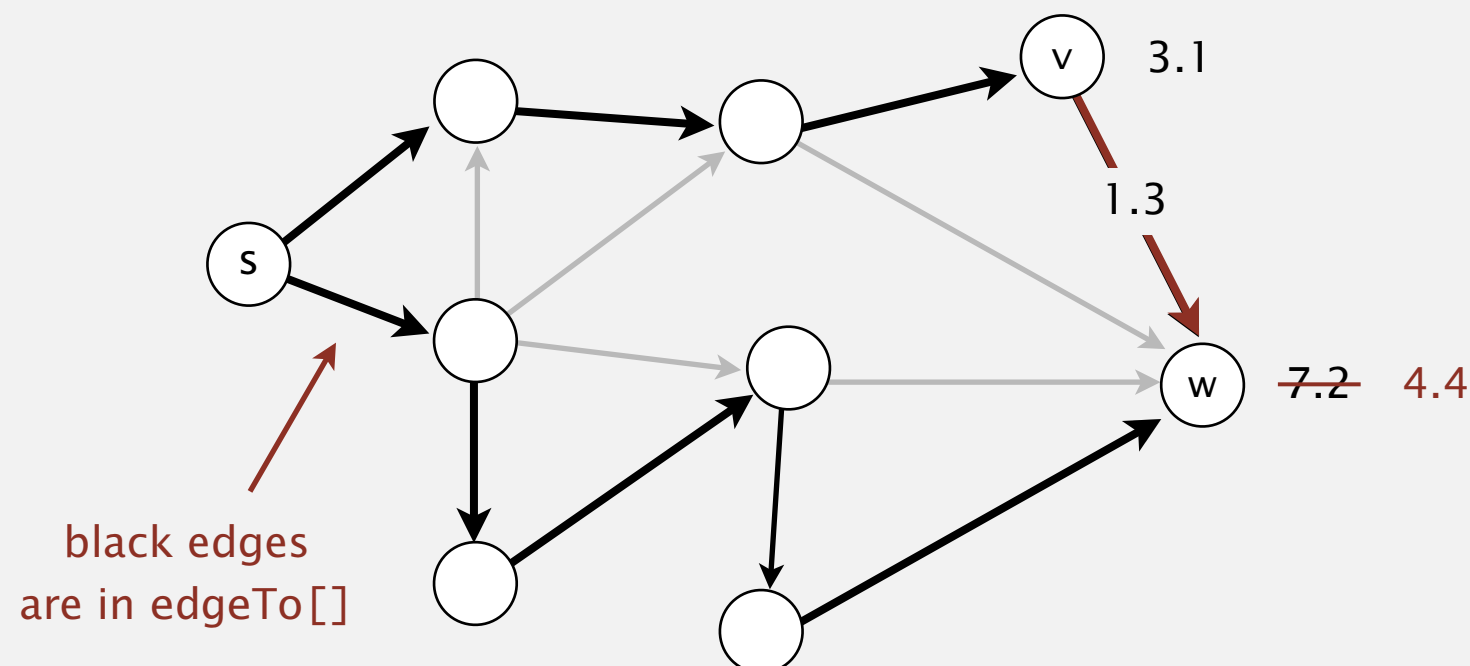
public Iterable<DirectedEdge> pathTo(int v)
{
    Stack<DirectedEdge> path = new Stack<DirectedEdge>();
    for (DirectedEdge e = edgeTo[v]; e != null; e = edgeTo[e.from()])
        path.push(e);
    return path;
}
```

Edge relaxation

Relax edge $e = v \rightarrow w$.

- $\text{distTo}[v]$ is length of shortest **known** path from s to v .
- $\text{distTo}[w]$ is length of shortest **known** path from s to w .
- $\text{edgeTo}[w]$ is last edge on shortest **known** path from s to w .
- If $e = v \rightarrow w$ gives shorter path to w through v , update both $\text{distTo}[w]$ and $\text{edgeTo}[w]$.

$v \rightarrow w$ successfully relaxes



Edge relaxation

Relax edge $e = v \rightarrow w$.

- `distTo[v]` is length of shortest **known** path from s to v .
- `distTo[w]` is length of shortest **known** path from s to w .
- `edgeTo[w]` is last edge on shortest **known** path from s to w .
- If $e = v \rightarrow w$ gives shorter path to w through v ,
update both `distTo[w]` and `edgeTo[w]`.

```
private void relax(DirectedEdge e)
{
    int v = e.from(), w = e.to();
    if (distTo[w] > distTo[v] + e.weight())
    {
        distTo[w] = distTo[v] + e.weight();
        edgeTo[w] = e;
    }
}
```

Shortest-paths optimality conditions

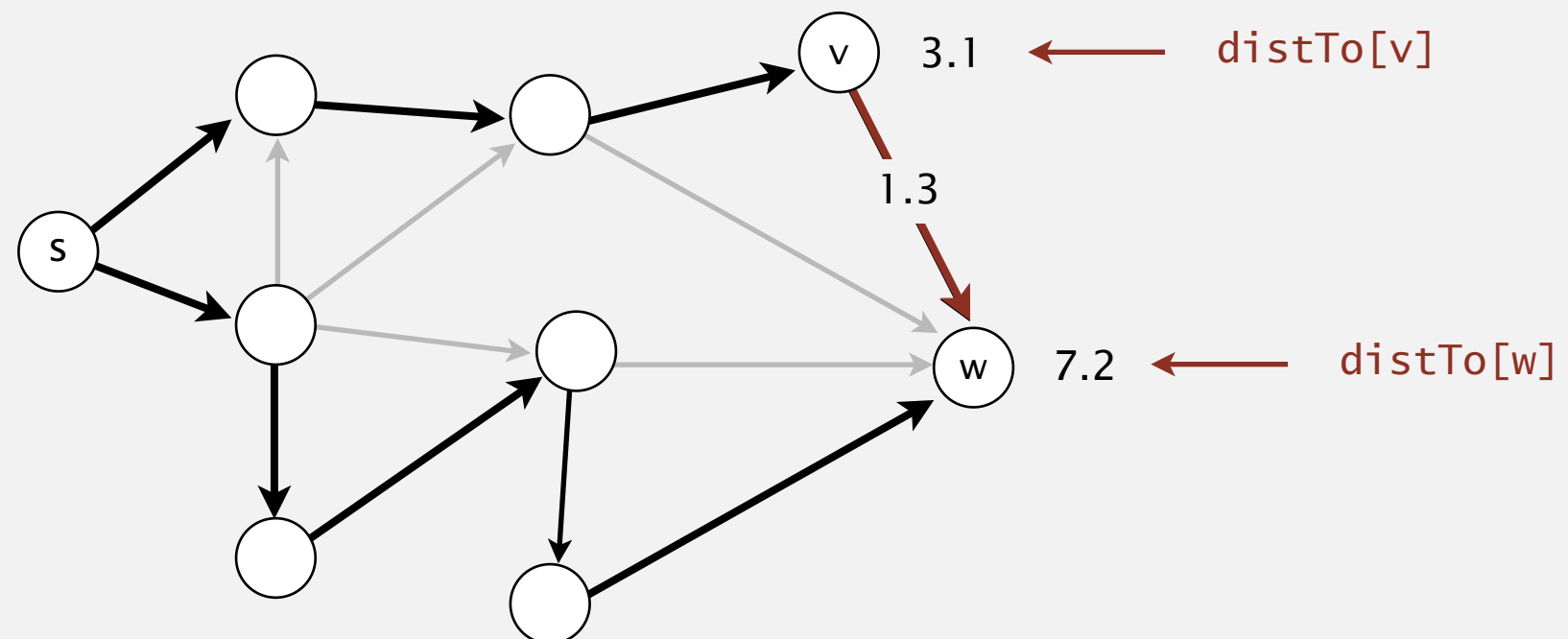
Proposition. Let G be an edge-weighted digraph.

Then $\text{distTo}[]$ are the shortest path distances from s iff:

- $\text{distTo}[s] = 0$.
- For each vertex v , $\text{distTo}[v]$ is the length of some path from s to v .
- For each edge $e = v \rightarrow w$, $\text{distTo}[w] \leq \text{distTo}[v] + e.\text{weight}()$.

Pf. $\Leftarrow$ [necessary]

- Suppose that $\text{distTo}[w] > \text{distTo}[v] + e.\text{weight}()$ for some edge $e = v \rightarrow w$.
- Then, e gives a path from s to w (through v) of length less than $\text{distTo}[w]$.



Shortest-paths optimality conditions

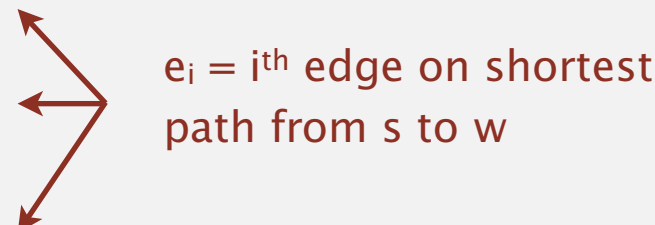
Proposition. Let G be an edge-weighted digraph.

Then $\text{distTo}[]$ are the shortest path distances from s iff:

- $\text{distTo}[s] = 0$.
- For each vertex v , $\text{distTo}[v]$ is the length of some path from s to v .
- For each edge $e = v \rightarrow w$, $\text{distTo}[w] \leq \text{distTo}[v] + e.\text{weight}()$.

Pf. $\Rightarrow$ [sufficient]

- Suppose that $s = v_0 \rightarrow v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_k = w$ is a shortest path from s to w .

- Then,
$$\begin{array}{llll} \text{distTo}[v_1] & \leq & \text{distTo}[v_0] & + e_1.\text{weight}() \\ \text{distTo}[v_2] & \leq & \text{distTo}[v_1] & + e_2.\text{weight}() \\ \dots & & & \\ \text{distTo}[v_k] & \leq & \text{distTo}[v_{k-1}] & + e_k.\text{weight}() \end{array}$$


- Add inequalities; simplify; and substitute $\text{distTo}[v_0] = \text{distTo}[s] = 0$:

$$\text{distTo}[w] = \text{distTo}[v_k] \leq \underline{e_1.\text{weight}() + e_2.\text{weight}() + \dots + e_k.\text{weight}()}$$

weight of shortest path from s to w

- Thus, $\text{distTo}[w]$ is the weight of shortest path to w . ■

 weight of some path from s to w

Generic shortest-paths algorithm

Generic algorithm (to compute SPT from s)

Initialize $\text{distTo}[s] = 0$ and $\text{distTo}[v] = \infty$ for all other vertices.

Repeat until optimality conditions are satisfied:

- Relax any edge.
-

Proposition. Generic algorithm computes SPT (if it exists) from s .

Pf sketch.

- The entry $\text{distTo}[v]$ is always the length of a simple path from s to v .
- Each successful relaxation decreases $\text{distTo}[v]$ for some v .
- The entry $\text{distTo}[v]$ can decrease at most a finite number of times. ■

Generic shortest-paths algorithm

Generic algorithm (to compute SPT from s)

Initialize $\text{distTo}[s] = 0$ and $\text{distTo}[v] = \infty$ for all other vertices.

Repeat until optimality conditions are satisfied:

- Relax any edge.
-

Efficient implementations. How to choose which edge to relax?

Ex 1. Dijkstra's algorithm (nonnegative weights).

Ex 2. Topological sort algorithm (no directed cycles).

Ex 3. Bellman-Ford algorithm (no negative cycles).



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Edsger W. Dijkstra: select quotes

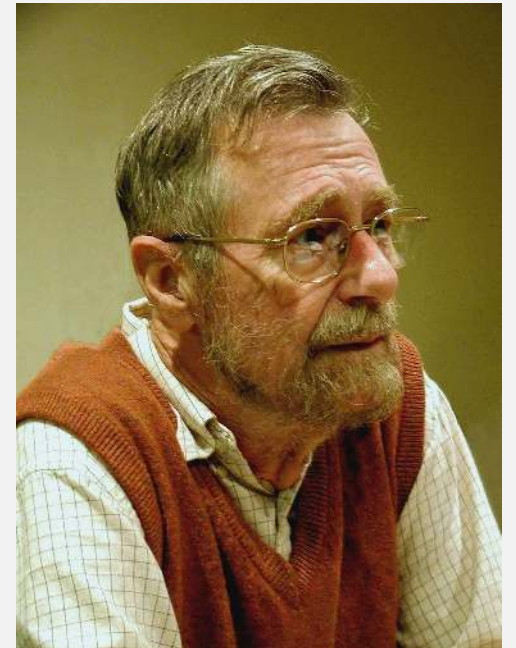
“ Do only what only you can do. ”

“ In their capacity as a tool, computers will be but a ripple on the surface of our culture. In their capacity as intellectual challenge, they are without precedent in the cultural history of mankind. ”

“ The use of COBOL cripples the mind; its teaching should, therefore, be regarded as a criminal offence. ”

“ It is practically impossible to teach good programming to students that have had a prior exposure to BASIC: as potential programmers they are mentally mutilated beyond hope of regeneration. ”

“ APL is a mistake, carried through to perfection. It is the language of the future for the programming techniques of the past: it creates a new generation of coding bums. ”



Edsger W. Dijkstra
Turing award 1972

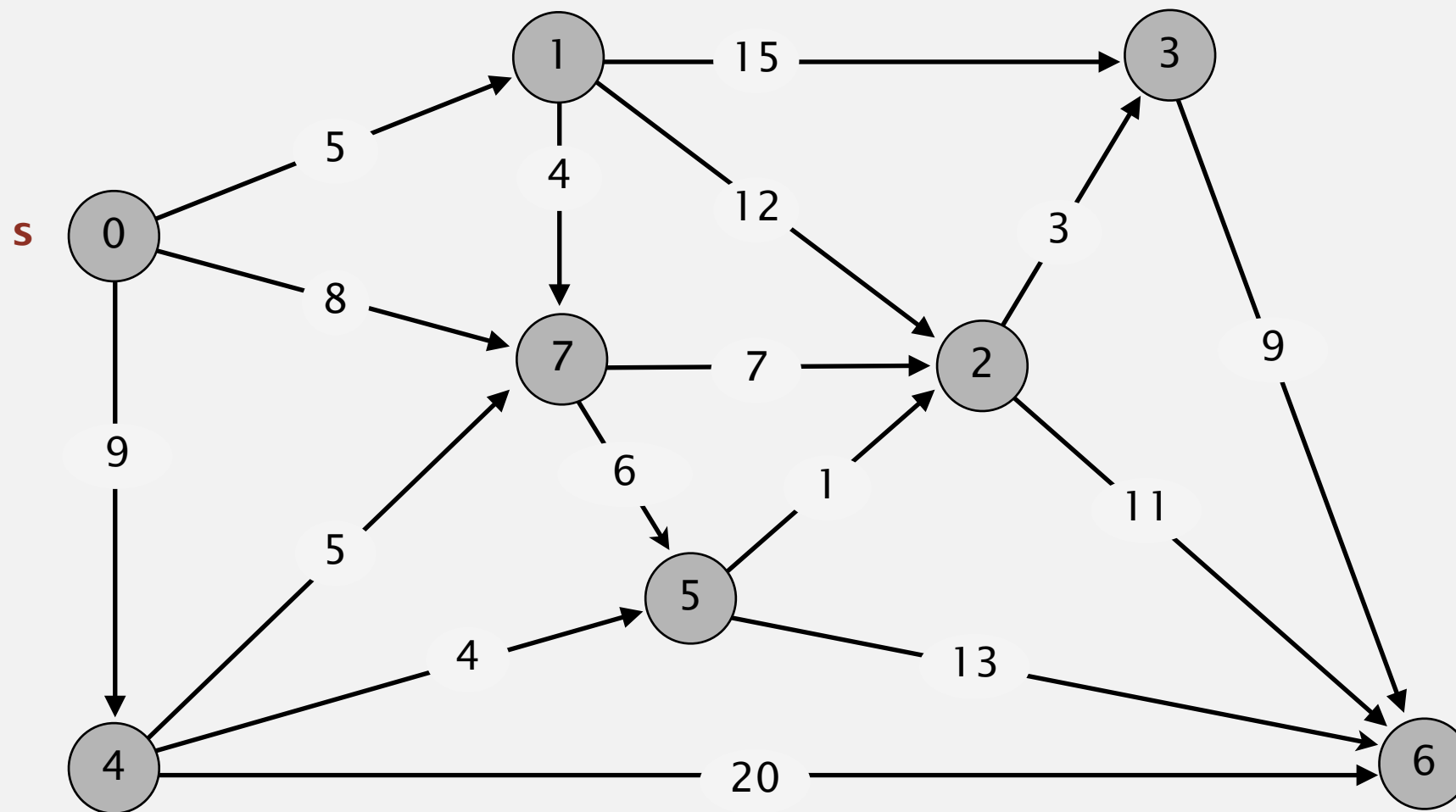
Edsger W. Dijkstra: select quotes



Dijkstra's algorithm demo



- Consider vertices in increasing order of distance from s (non-tree vertex with the lowest $\text{distTo}[]$ value).
- Add vertex to tree and relax all edges pointing from that vertex.

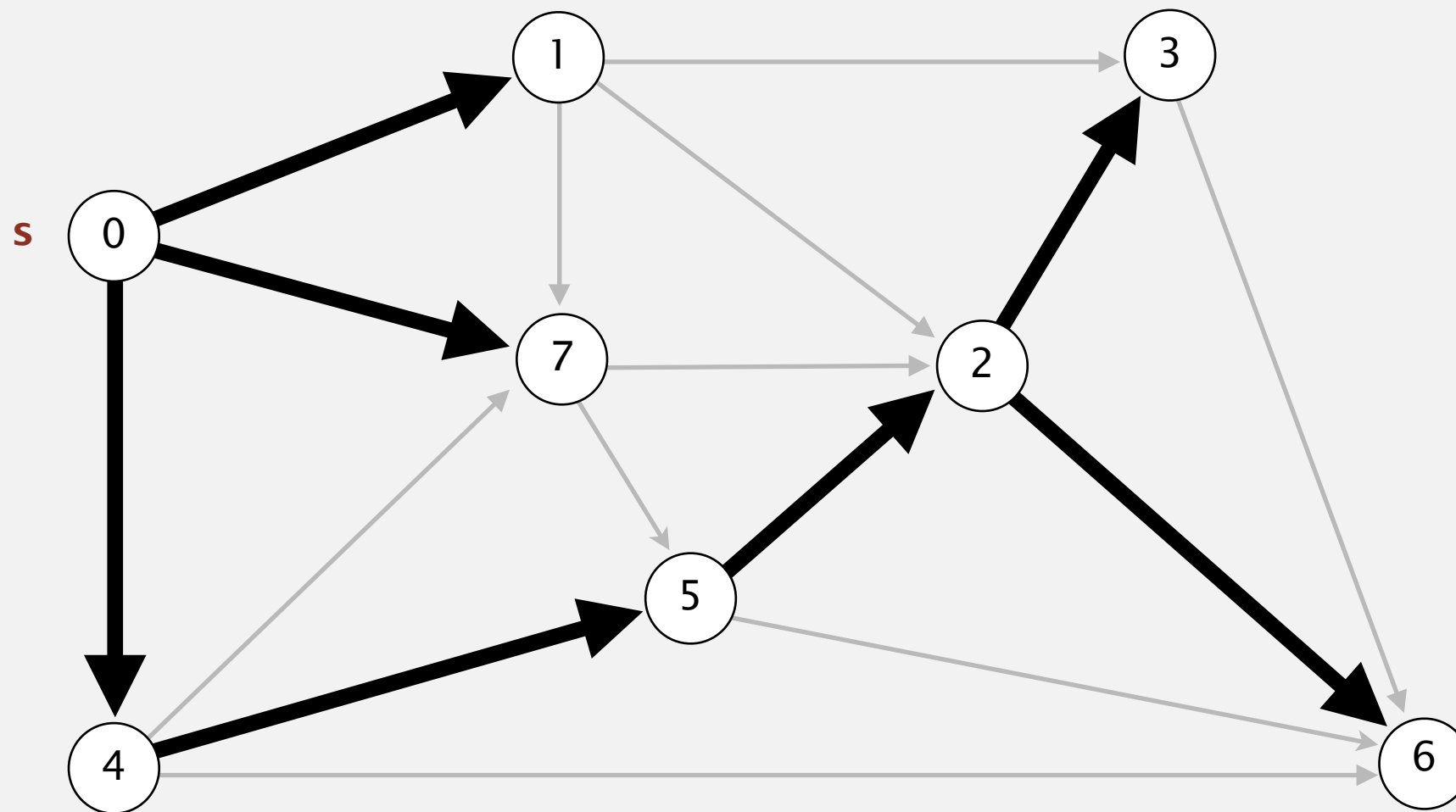


an edge-weighted digraph

0→1	5.0
0→4	9.0
0→7	8.0
1→2	12.0
1→3	15.0
1→7	4.0
2→3	3.0
2→6	11.0
3→6	9.0
4→5	4.0
4→6	20.0
4→7	5.0
5→2	1.0
5→6	13.0
7→5	6.0
7→2	7.0

Dijkstra's algorithm demo

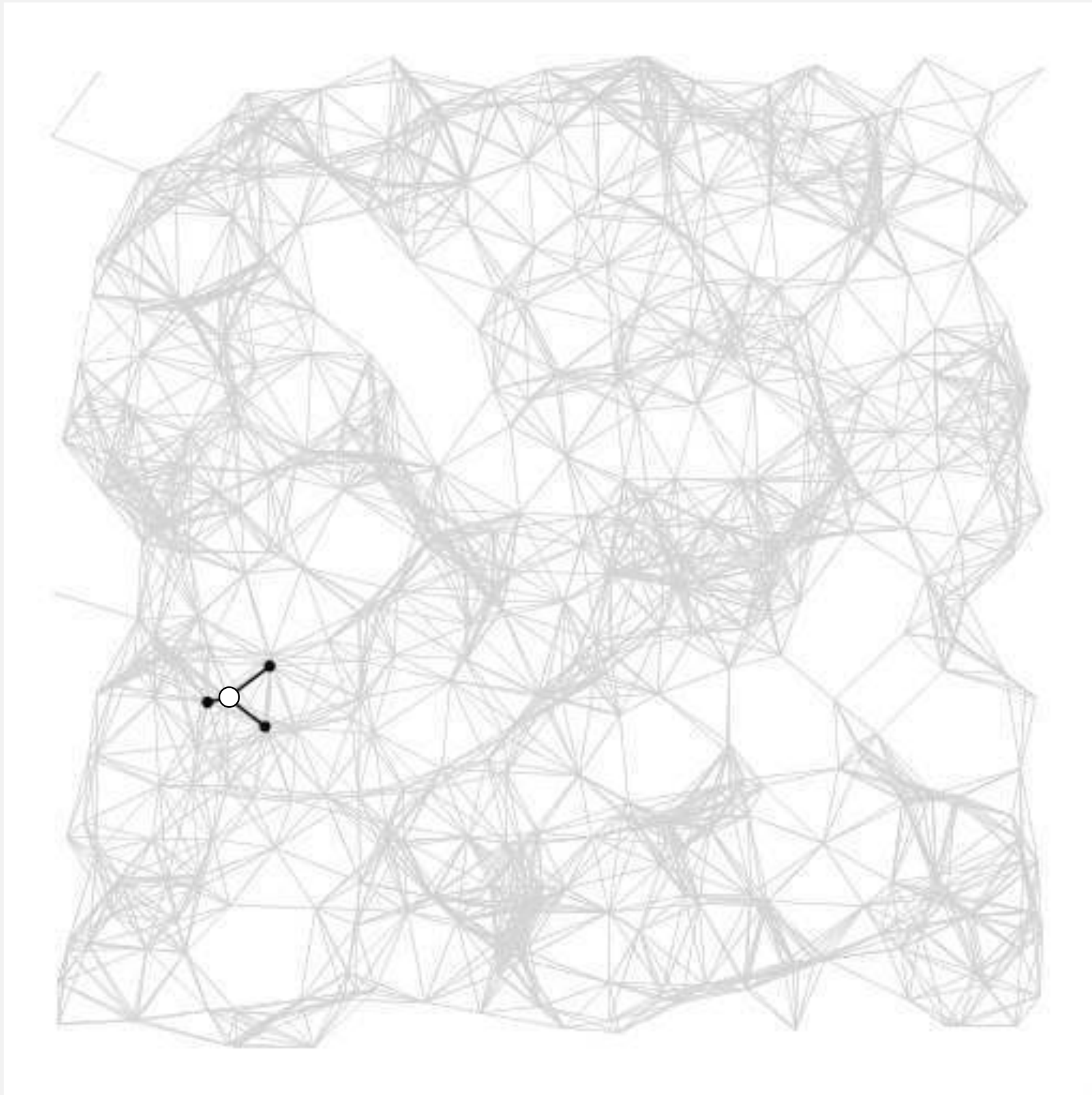
- Consider vertices in increasing order of distance from s (non-tree vertex with the lowest `distTo[]` value).
- Add vertex to tree and relax all edges pointing from that vertex.



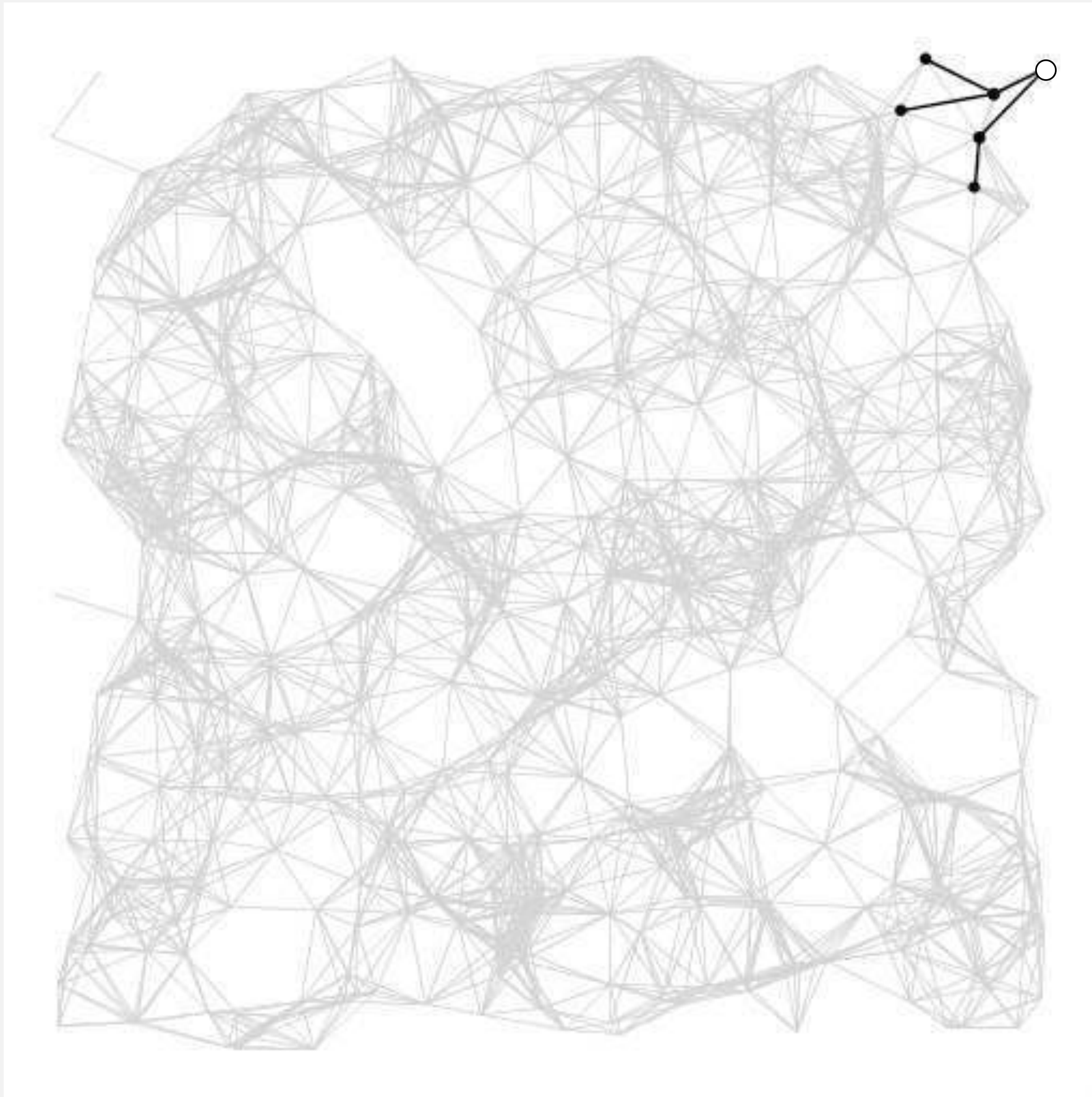
v	distTo[]	edgeTo[]
0	0.0	-
1	5.0	0→1
2	14.0	5→2
3	17.0	2→3
4	9.0	0→4
5	13.0	4→5
6	25.0	2→6
7	8.0	0→7

shortest-paths tree from vertex s

Dijkstra's algorithm visualization



Dijkstra's algorithm visualization



Dijkstra's algorithm: correctness proof

Proposition. Dijkstra's algorithm computes a SPT in any edge-weighted digraph with nonnegative weights.

Pf.

- Each edge $e = v \rightarrow w$ is relaxed exactly once (when vertex v is relaxed), leaving $\text{distTo}[w] \leq \text{distTo}[v] + e.\text{weight}()$.
- Inequality holds until algorithm terminates because:
 - $\text{distTo}[w]$ cannot increase  $\text{distTo}[]$ values are monotone decreasing
 - $\text{distTo}[v]$ will not change  we choose lowest $\text{distTo}[]$ value at each step (and edge weights are nonnegative)
- Thus, upon termination, shortest-paths optimality conditions hold. ■

Dijkstra's algorithm: Java implementation

```
public class DijkstraSP
{
    private DirectedEdge[] edgeTo;
    private double[] distTo;
    private IndexMinPQ<Double> pq;

    public DijkstraSP(EdgeWeightedDigraph G, int s)
    {
        edgeTo = new DirectedEdge[G.V()];
        distTo = new double[G.V()];
        pq = new IndexMinPQ<Double>(G.V());

        for (int v = 0; v < G.V(); v++)
            distTo[v] = Double.POSITIVE_INFINITY;
        distTo[s] = 0.0;

        pq.insert(s, 0.0);
        while (!pq.isEmpty())
        {
            int v = pq.delMin();
            for (DirectedEdge e : G.adj(v))
                relax(e);
        }
    }
}
```

← relax vertices in order
of distance from s

Dijkstra's algorithm: Java implementation

```
private void relax(DirectedEdge e)
{
    int v = e.from(), w = e.to();
    if (distTo[w] > distTo[v] + e.weight())
    {
        distTo[w] = distTo[v] + e.weight();
        edgeTo[w] = e;
        if (pq.contains(w)) pq.decreaseKey(w, distTo[w]);
        else                pq.insert      (w, distTo[w]);
    }
}
```

← update PQ

Dijkstra's algorithm: which priority queue?

Depends on PQ implementation: V insert, V delete-min, E decrease-key.

PQ implementation	insert	delete-min	decrease-key	total
unordered array	1	V	1	V^2
binary heap	$\log V$	$\log V$	$\log V$	$E \log V$
d-way heap	$\log_d V$	$d \log_d V$	$\log_d V$	$E \log_{E/V} V$
Fibonacci heap	$1^\dagger$	$\log V^\dagger$	$1^\dagger$	$E + V \log V$

† amortized

Bottom line.

- Array implementation optimal for dense graphs.
- Binary heap much faster for sparse graphs.
- 4-way heap worth the trouble in performance-critical situations.
- Fibonacci heap best in theory, but not worth implementing.

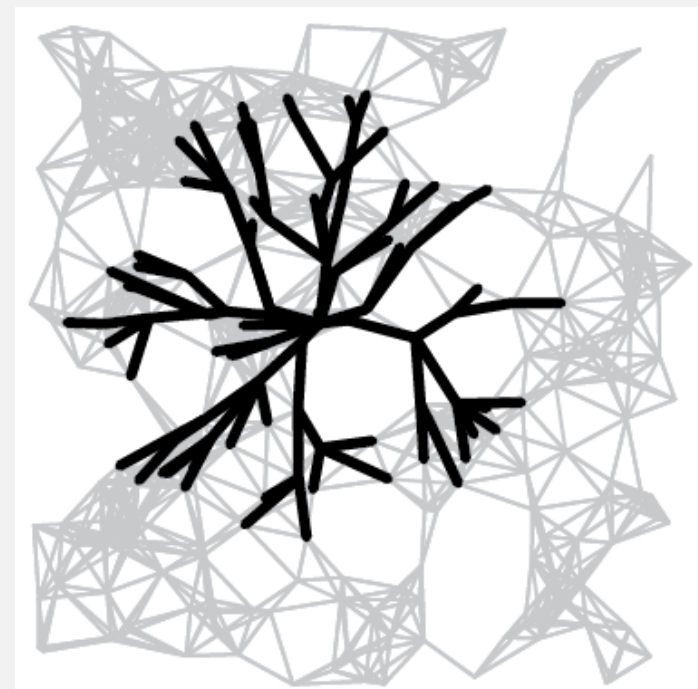
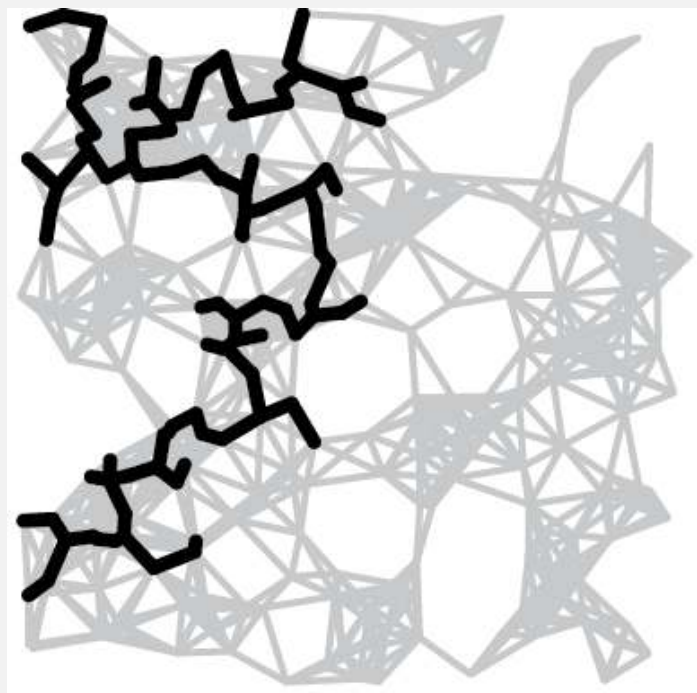
Computing a spanning tree in a graph

Dijkstra's algorithm seem familiar?

- Prim's algorithm is essentially the same algorithm.
- Both are in a family of algorithms that compute a spanning tree.

Main distinction: Rule used to choose next vertex for the tree.

- Prim: Closest vertex to the **tree** (via an undirected edge).
- Dijkstra: Closest vertex to the **source** (via a directed path).



Note: DFS and BFS are also in this family of algorithms.



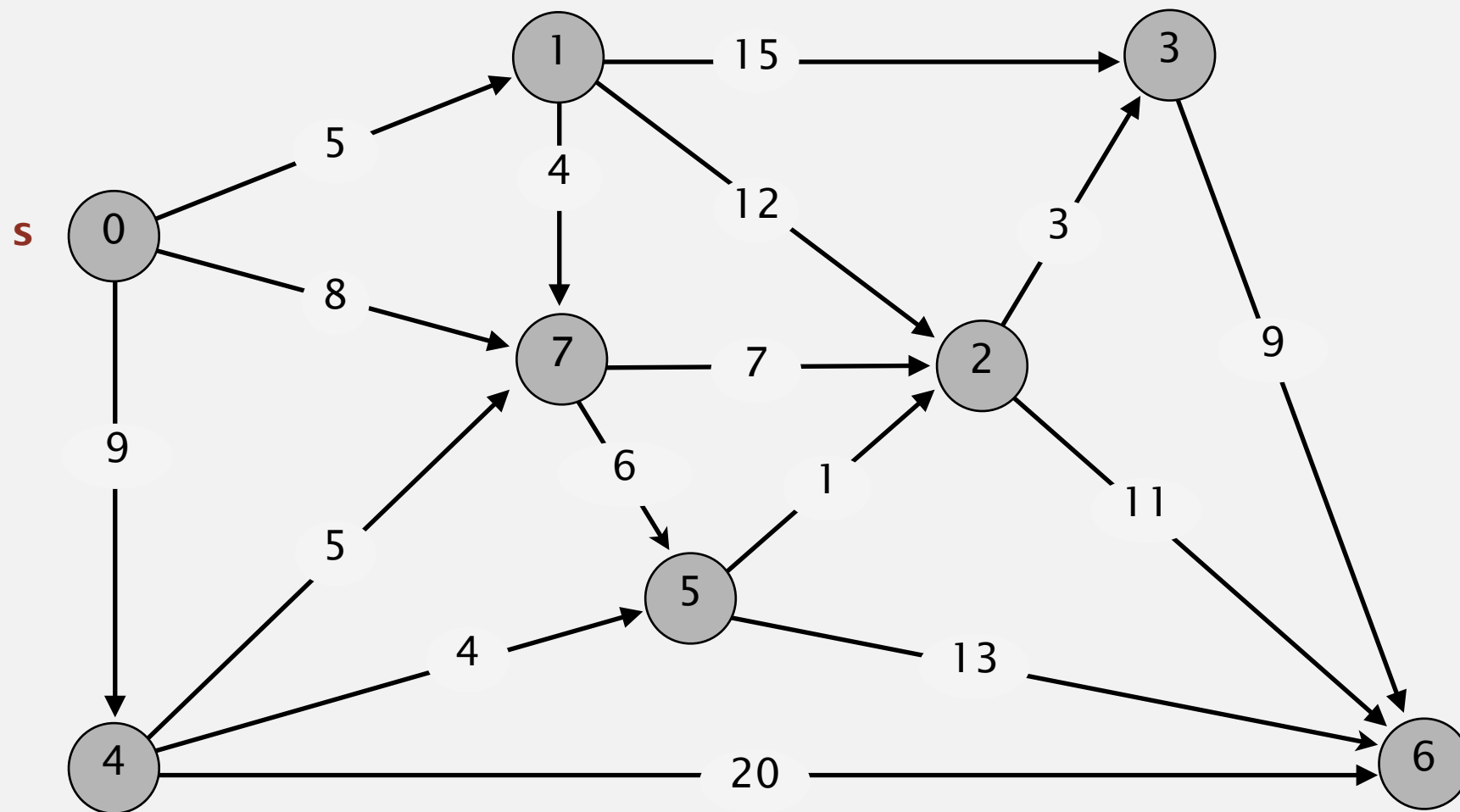
<http://algs4.cs.princeton.edu>

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Acyclic edge-weighted digraphs

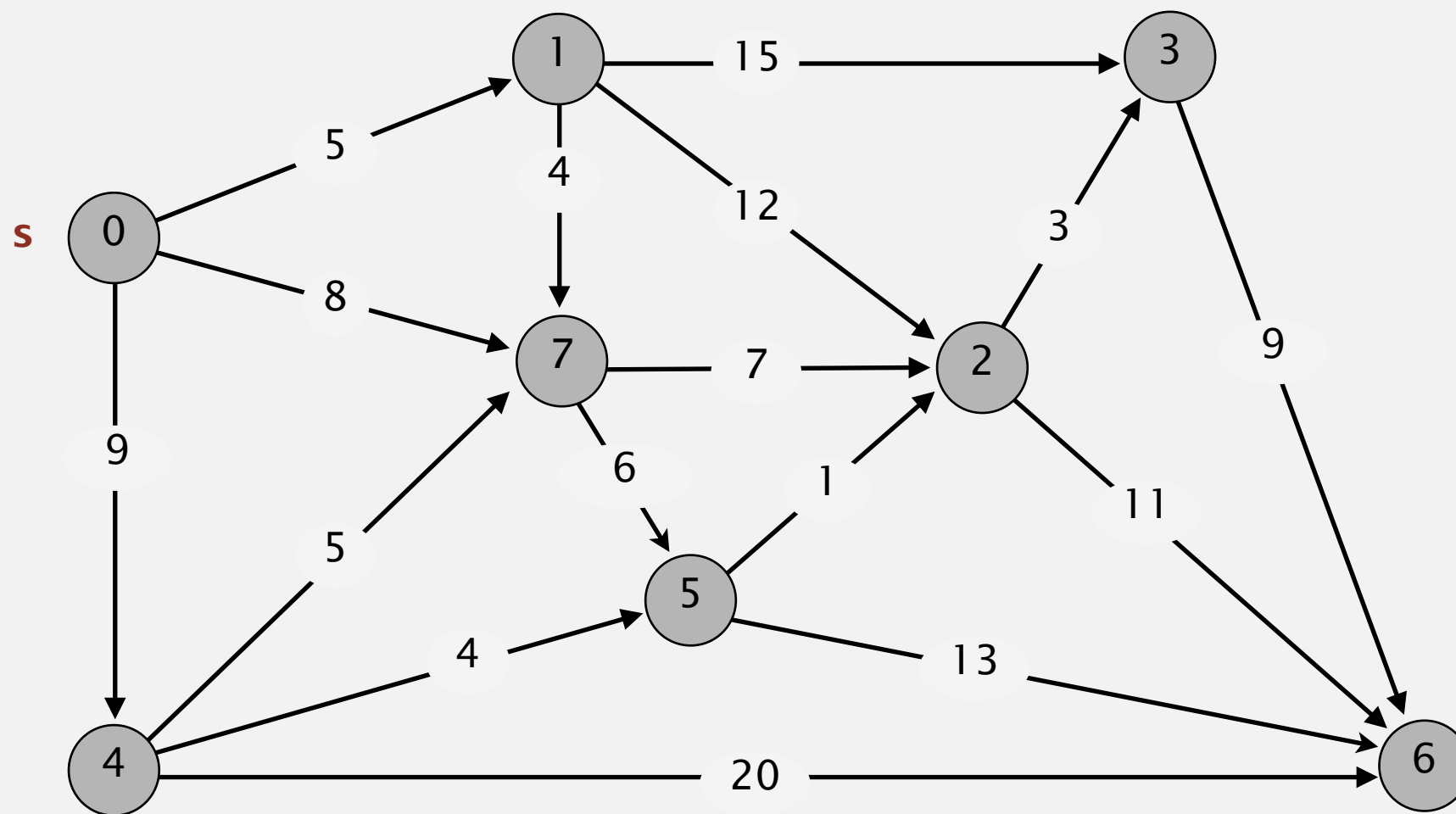
Q. Suppose that an edge-weighted digraph has no directed cycles. Is it easier to find shortest paths than in a general digraph?



A. Yes!

Acyclic shortest paths demo

- Consider vertices in topological order.
- Relax all edges pointing from that vertex.

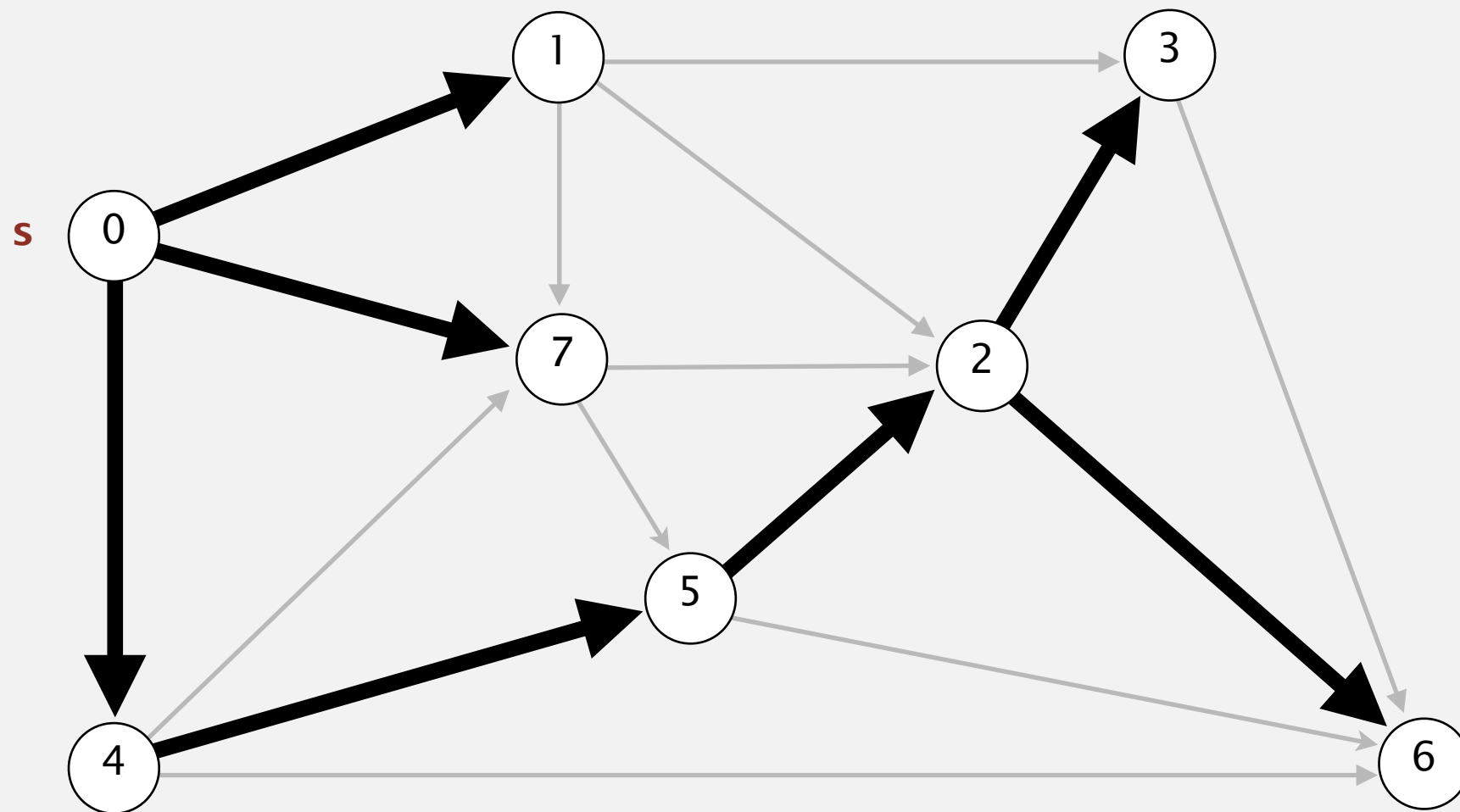


0→1	5.0
0→4	9.0
0→7	8.0
1→2	12.0
1→3	15.0
1→7	4.0
2→3	3.0
2→6	11.0
3→6	9.0
4→5	4.0
4→6	20.0
4→7	5.0
5→2	1.0
5→6	13.0
7→5	6.0
7→2	7.0

an edge-weighted DAG

Acyclic shortest paths demo

- Consider vertices in topological order.
- Relax all edges pointing from that vertex.



shortest-paths tree from vertex s

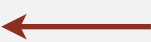
	0	1	4	7	5	2	3	6
v	distTo[]		edgeTo[]					
0	0.0		-					
1	5.0		0→1					
2	14.0		5→2					
3	17.0		2→3					
4	9.0		0→4					
5	13.0		4→5					
6	25.0		2→6					
7	8.0		0→7					

Shortest paths in edge-weighted DAGs

Proposition. Topological sort algorithm computes SPT in any edge-weighted DAG in time proportional to $E + V$.

edge weights
can be negative!

Pf.

- Each edge $e = v \rightarrow w$ is relaxed exactly once (when vertex v is relaxed), leaving $\text{distTo}[w] \leq \text{distTo}[v] + e.\text{weight}()$.
- Inequality holds until algorithm terminates because:
 - $\text{distTo}[w]$ cannot increase  $\text{distTo}[]$ values are monotone decreasing
 - $\text{distTo}[v]$ will not change  because of topological order, no edge pointing to v will be relaxed after v is relaxed
- Thus, upon termination, shortest-paths optimality conditions hold. ■

Shortest paths in edge-weighted DAGs

```
public class AcyclicSP
{
    private DirectedEdge[] edgeTo;
    private double[] distTo;

    public AcyclicSP(EdgeWeightedDigraph G, int s)
    {
        edgeTo = new DirectedEdge[G.V()];
        distTo = new double[G.V()];

        for (int v = 0; v < G.V(); v++)
            distTo[v] = Double.POSITIVE_INFINITY;
        distTo[s] = 0.0;

        Topological topological = new Topological(G);
        for (int v : topological.order())
            for (DirectedEdge e : G.adj(v))
                relax(e);
    }
}
```

← topological order

Content-aware resizing

Seam carving. [Avidan and Shamir] Resize an image without distortion for display on cell phones and web browsers.



<http://www.youtube.com/watch?v=vIFCV2spKtg>

Content-aware resizing

Seam carving. [Avidan and Shamir] Resize an image without distortion for display on cell phones and web browsers.



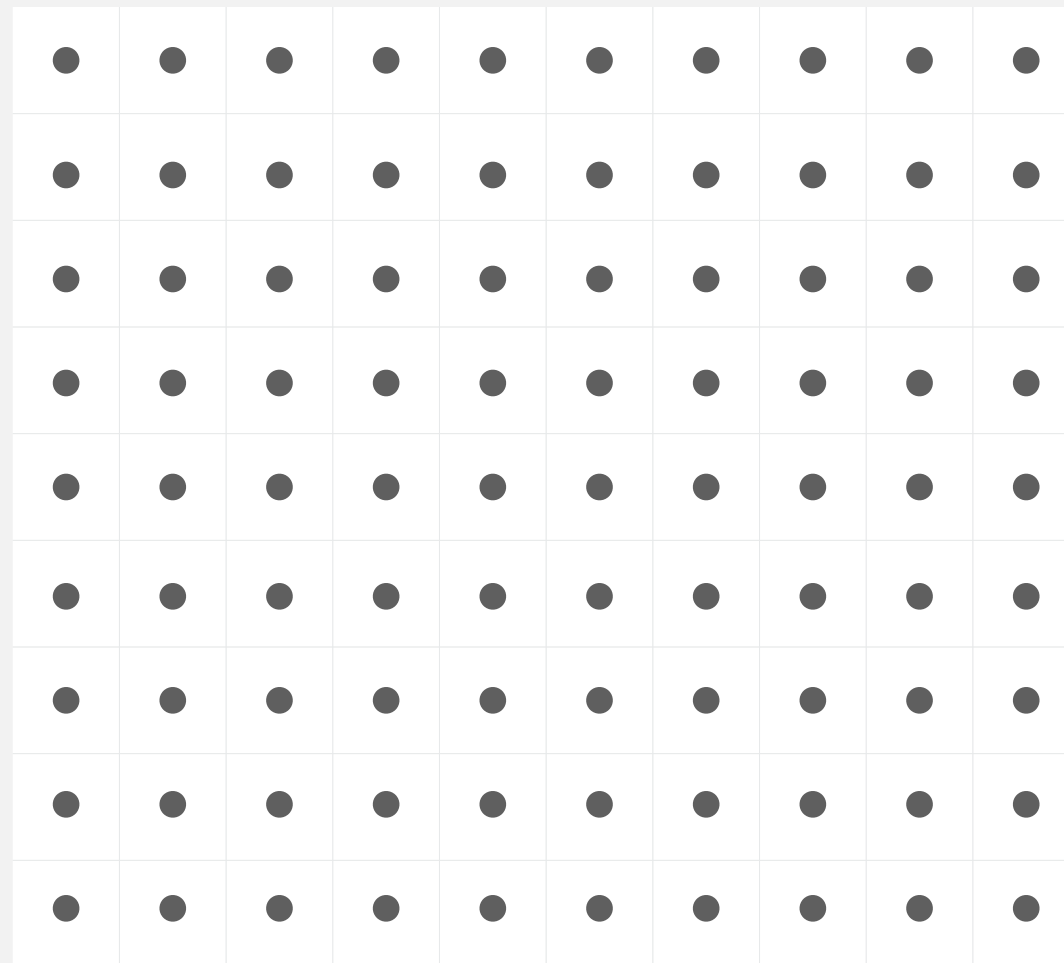
In the wild. Photoshop CS 5, Imagemagick, GIMP, ...



Content-aware resizing

To find vertical seam:

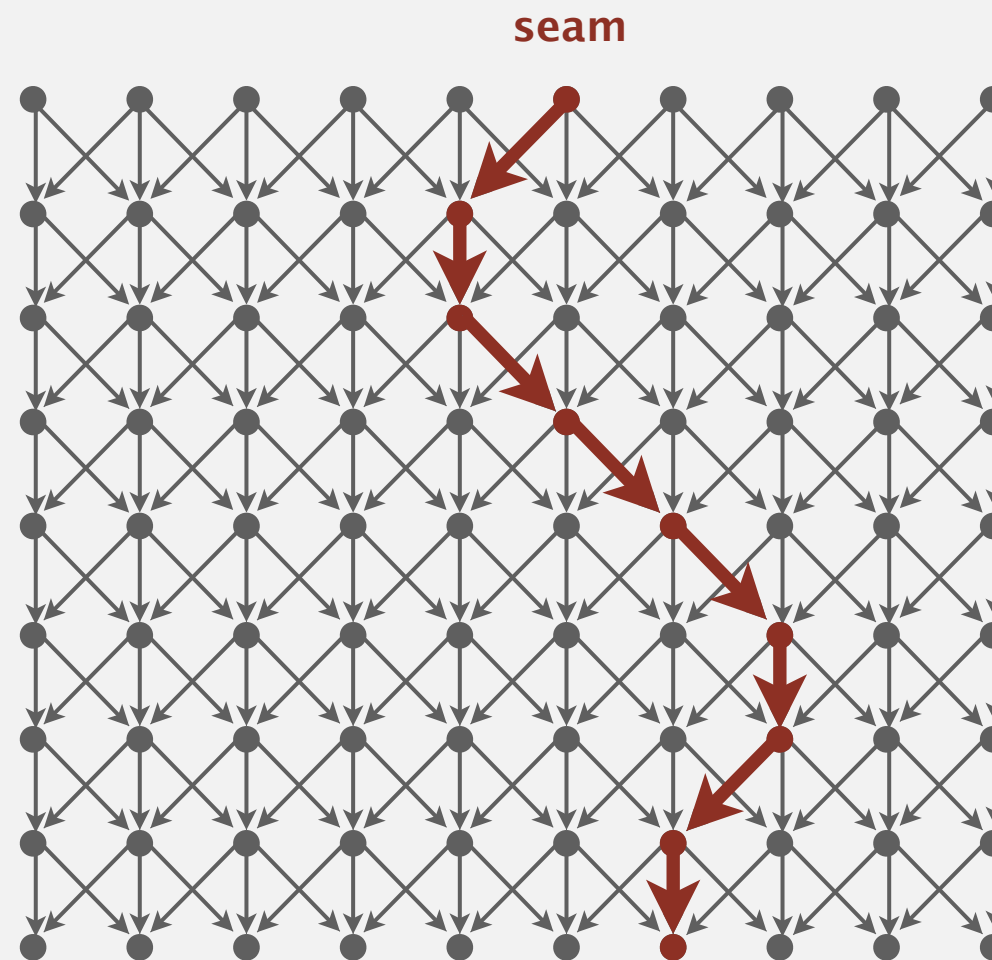
- Grid DAG: vertex = pixel; edge = from pixel to 3 downward neighbors.
- Weight of pixel = energy function of 8 neighboring pixels.
- Seam = shortest path (sum of vertex weights) from top to bottom.



Content-aware resizing

To find vertical seam:

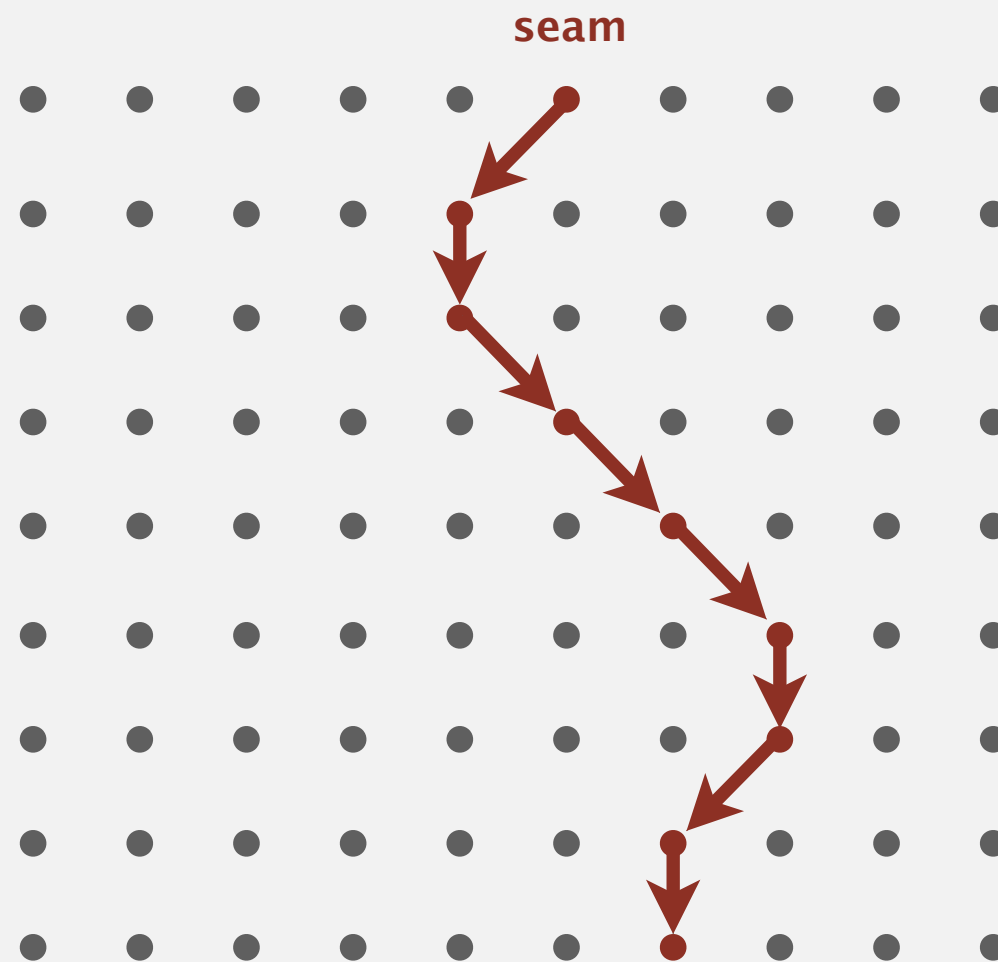
- Grid DAG: vertex = pixel; edge = from pixel to 3 downward neighbors.
- Weight of pixel = energy function of 8 neighboring pixels.
- Seam = shortest path (sum of vertex weights) from top to bottom.



Content-aware resizing

To remove vertical seam:

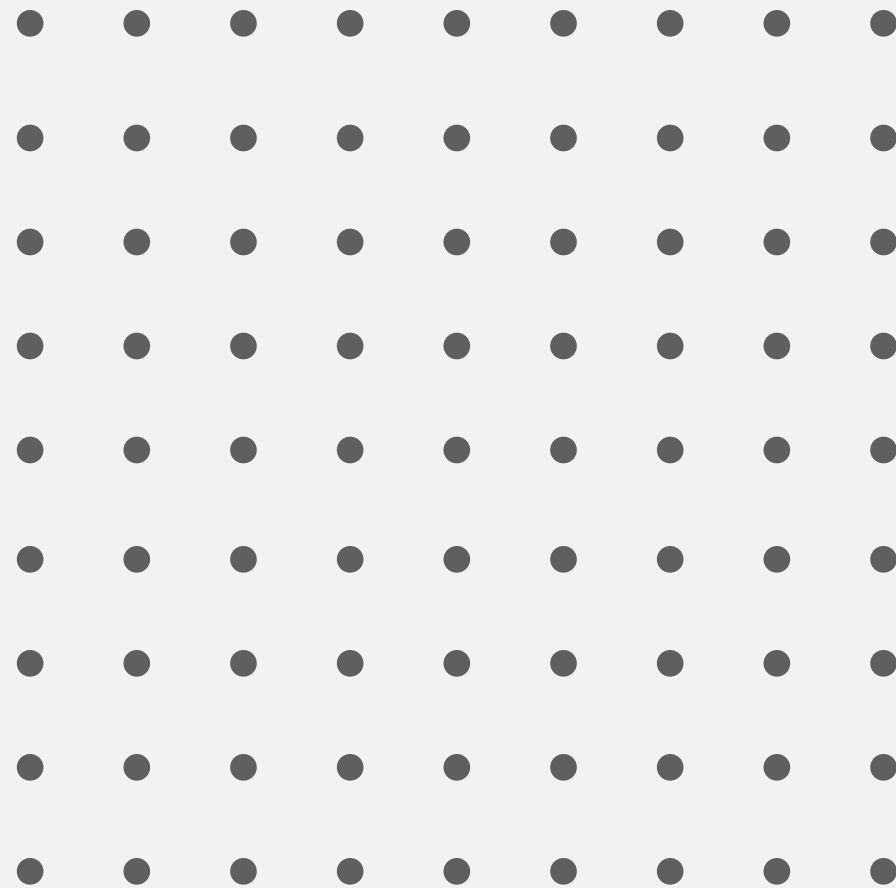
- Delete pixels on seam (one in each row).



Content-aware resizing

To remove vertical seam:

- Delete pixels on seam (one in each row).



Longest paths in edge-weighted DAGs

Formulate as a shortest paths problem in edge-weighted DAGs.

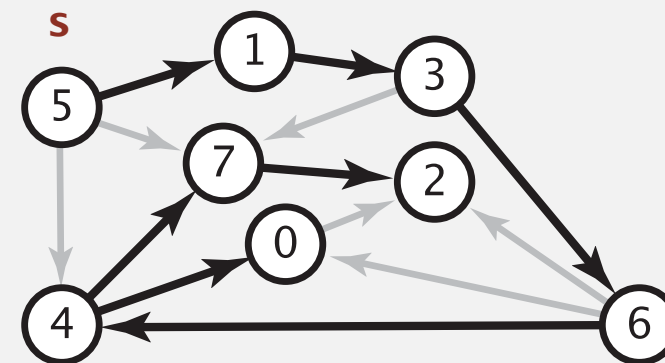
- Negate all weights.
 - Find shortest paths.
 - Negate weights in result.
- equivalent: reverse sense of equality in relax()

longest paths input

5→4 0.35
4→7 0.37
5→7 0.28
5→1 0.32
4→0 0.38
0→2 0.26
3→7 0.39
1→3 0.29
7→2 0.34
6→2 0.40
3→6 0.52
6→0 0.58
6→4 0.93

shortest paths input

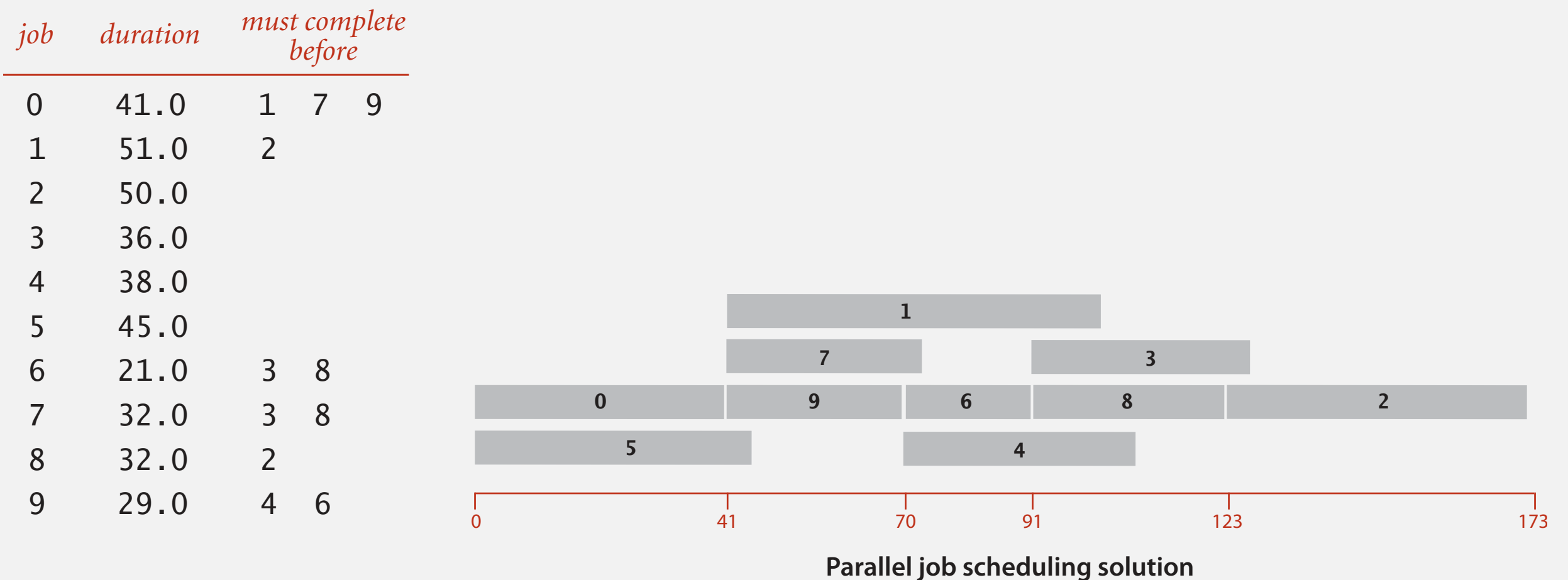
5→4 -0.35
4→7 -0.37
5→7 -0.28
5→1 -0.32
4→0 -0.38
0→2 -0.26
3→7 -0.39
1→3 -0.29
7→2 -0.34
6→2 -0.40
3→6 -0.52
6→0 -0.58
6→4 -0.93



Key point. Topological sort algorithm works even with negative weights.

Longest paths in edge-weighted DAGs: application

Parallel job scheduling. Given a set of jobs with durations and precedence constraints, schedule the jobs (by finding a start time for each) so as to achieve the minimum completion time, while respecting the constraints.

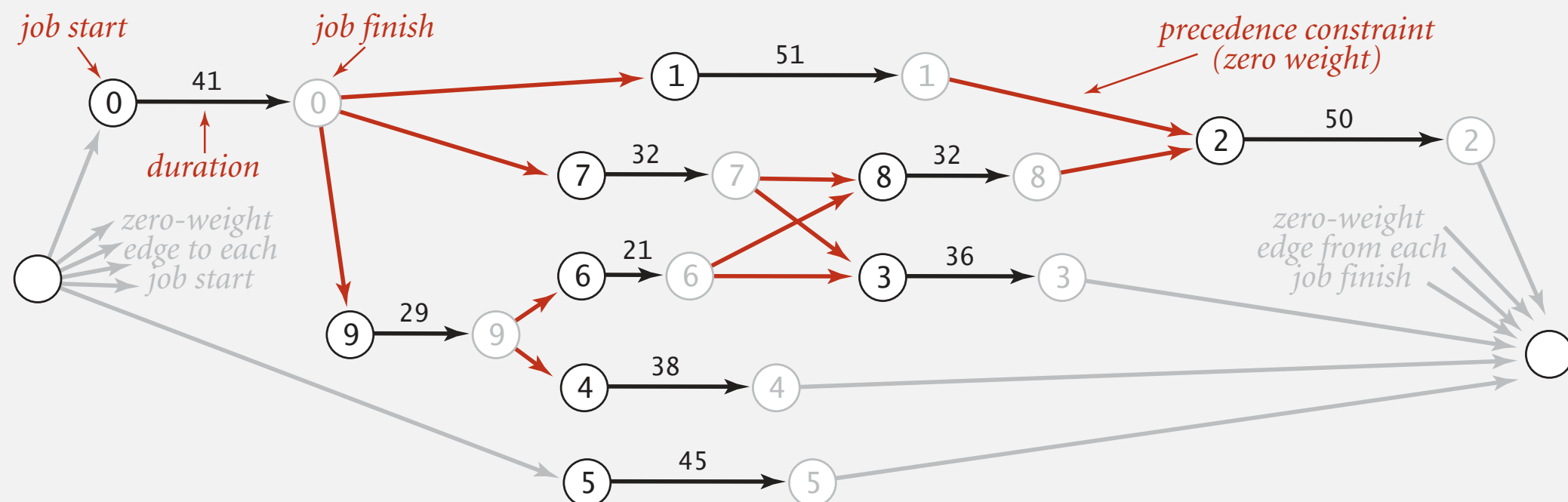


Critical path method

CPM. To solve a parallel job-scheduling problem, create edge-weighted DAG:

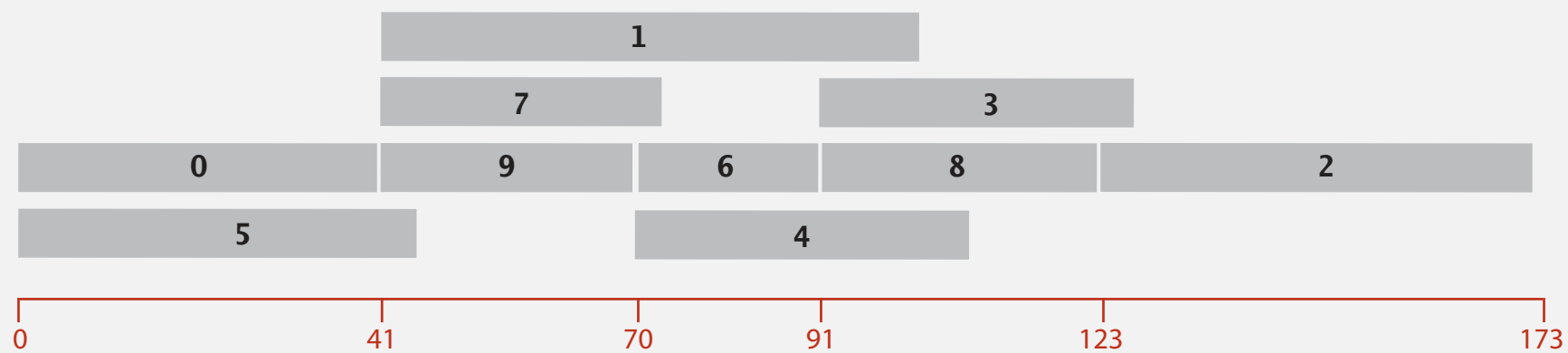
- Source and sink vertices.
- Two vertices (begin and end) for each job.
- Three edges for each job.
 - begin to end (weighted by duration)
 - source to begin (0 weight)
 - end to sink (0 weight)
- One edge for each precedence constraint (0 weight).

job	duration	must complete before		
0	41.0	1	7	9
1	51.0	2		
2	50.0			
3	36.0			
4	38.0			
5	45.0			
6	21.0	3	8	
7	32.0	3	8	
8	32.0	2		
9	29.0	4	6	

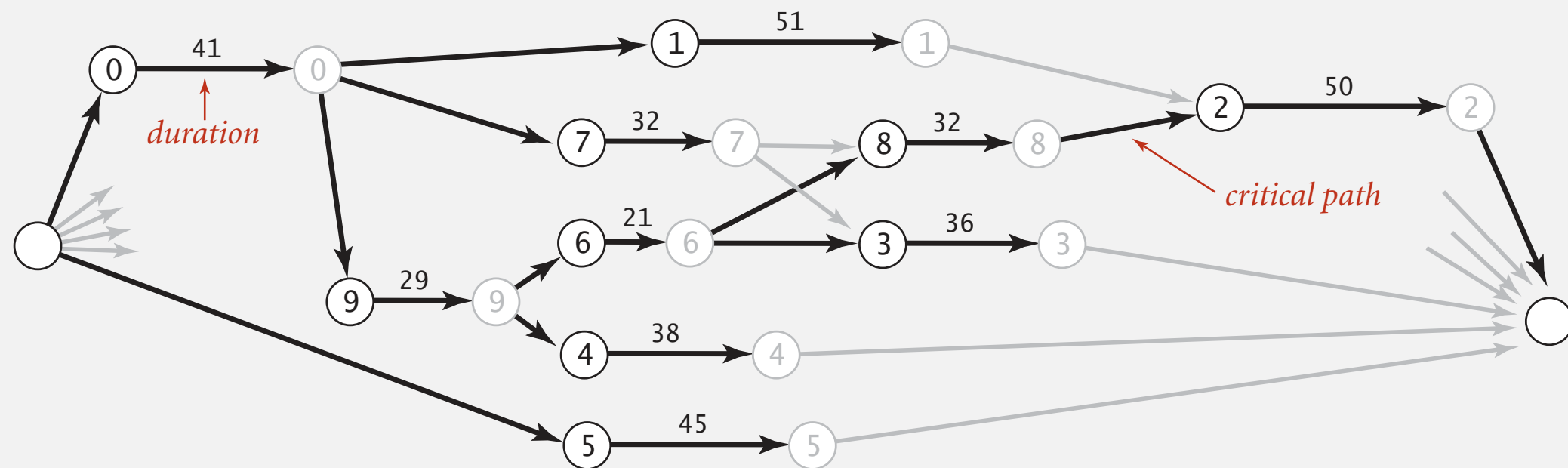


Critical path method

CPM. Use **longest path** from the source to schedule each job.



Parallel job scheduling solution





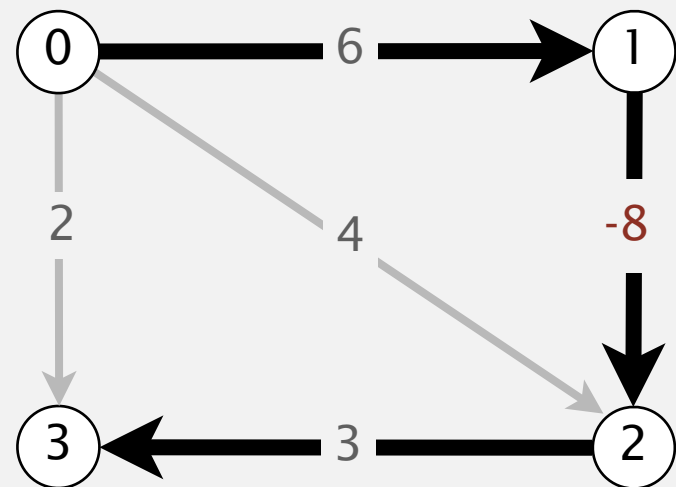
<http://algs4.cs.princeton.edu>

4.4 SHORTEST PATHS

- ▶ *APIs*
- ▶ *shortest-paths properties*
- ▶ *Dijkstra's algorithm*
- ▶ *edge-weighted DAGs*
- ▶ *negative weights*

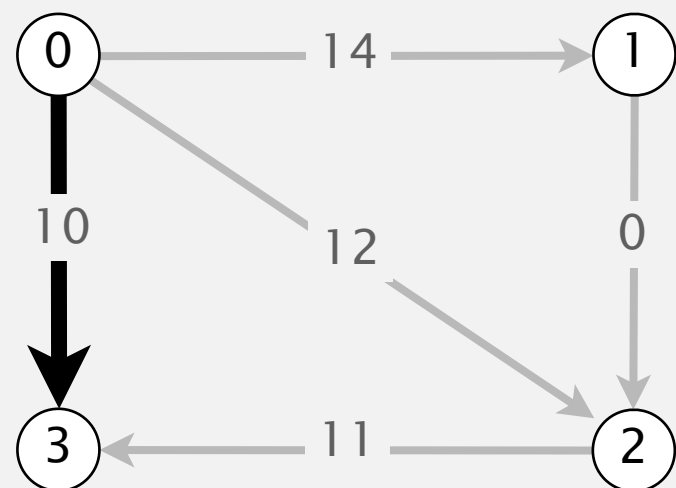
Shortest paths with negative weights: failed attempts

Dijkstra. Doesn't work with negative edge weights.



Dijkstra selects vertex 3 immediately after 0.
But shortest path from 0 to 3 is $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$.

Re-weighting. Add a constant to every edge weight doesn't work.



Adding 8 to each edge weight changes the
shortest path from $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$ to $0 \rightarrow 3$.

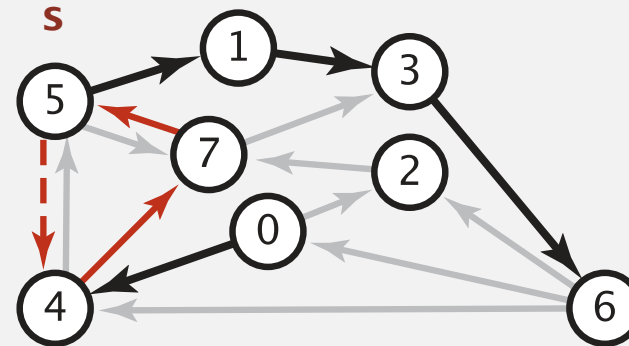
Conclusion. Need a different algorithm.

Negative cycles

Def. A **negative cycle** is a directed cycle whose sum of edge weights is negative.

digraph

4→5	0.35
5→4	-0.66
4→7	0.37
5→7	0.28
7→5	0.28
5→1	0.32
0→4	0.38
0→2	0.26
7→3	0.39
1→3	0.29
2→7	0.34
6→2	0.40
3→6	0.52
6→0	0.58
6→4	0.93



negative cycle $(-0.66 + 0.37 + 0.28)$

5→4→7→5

shortest path from 0 to 6

0→4→7→5→4→7→5...→1→3→6

Proposition. A SPT exists iff no negative cycles.

assuming all vertices reachable from s

Bellman-Ford algorithm

Bellman-Ford algorithm

Initialize $\text{distTo}[s] = 0$ and $\text{distTo}[v] = \infty$ for all other vertices.

Repeat V times:

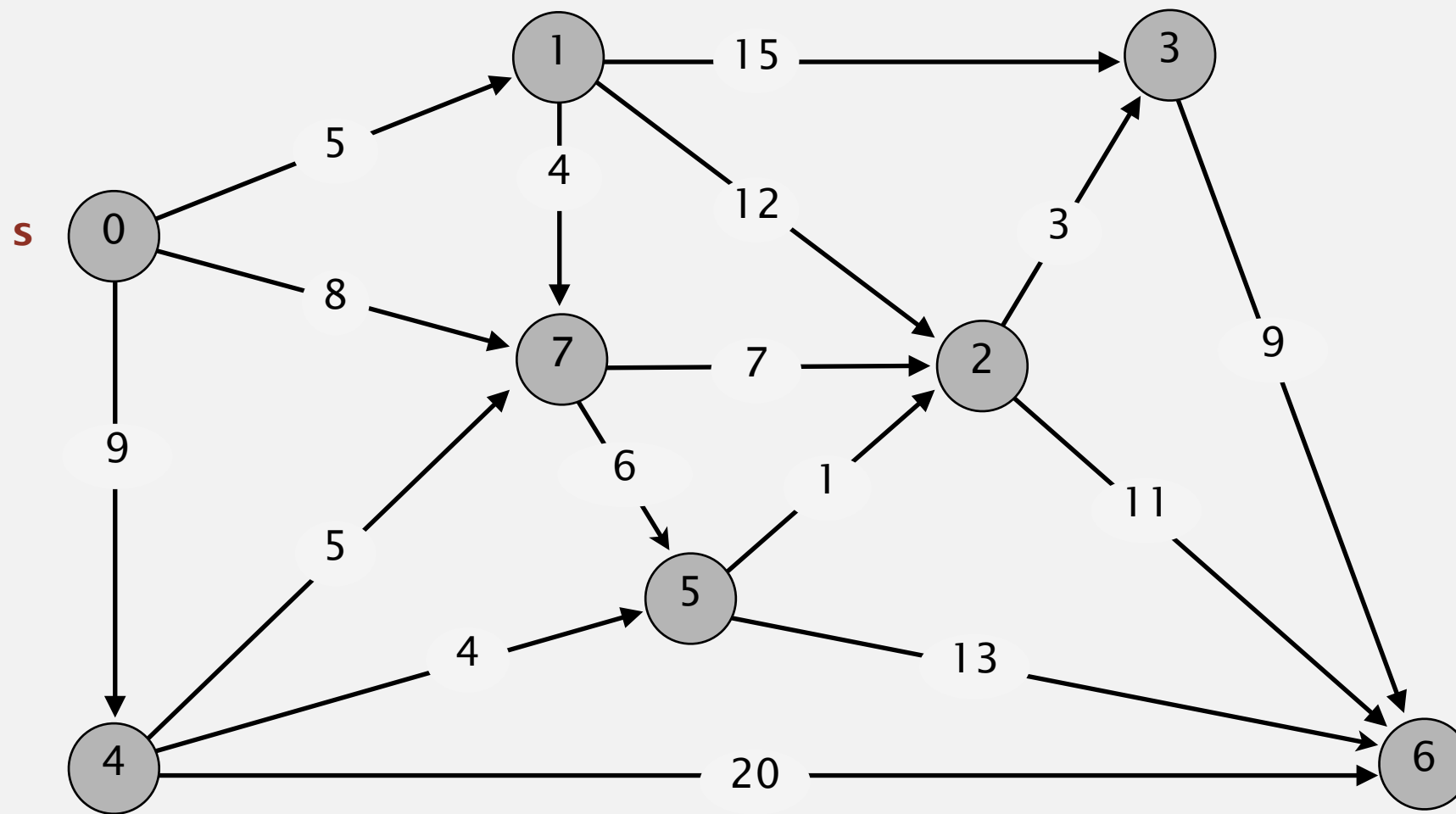
- Relax each edge.**
-

```
for (int i = 0; i < G.V(); i++)  
    for (int v = 0; v < G.V(); v++)  
        for (DirectedEdge e : G.adj(v))  
            relax(e);
```

← pass i (relax each edge)

Bellman-Ford algorithm demo

Repeat V times: relax all E edges.

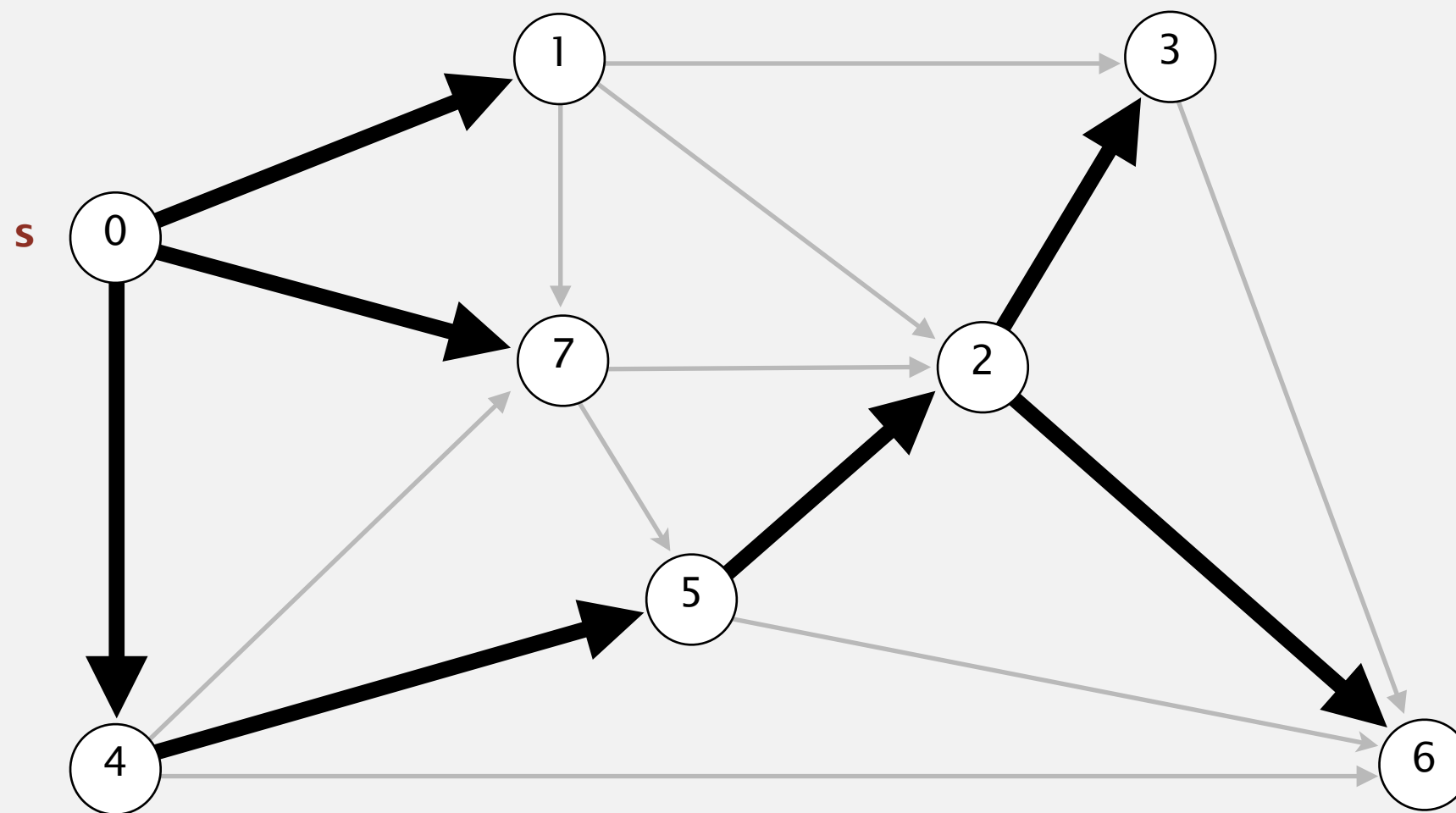


0→1	5.0
0→4	9.0
0→7	8.0
1→2	12.0
1→3	15.0
1→7	4.0
2→3	3.0
2→6	11.0
3→6	9.0
4→5	4.0
4→6	20.0
4→7	5.0
5→2	1.0
5→6	13.0
7→5	6.0
7→2	7.0

an edge-weighted digraph

Bellman-Ford algorithm demo

Repeat V times: relax all E edges.

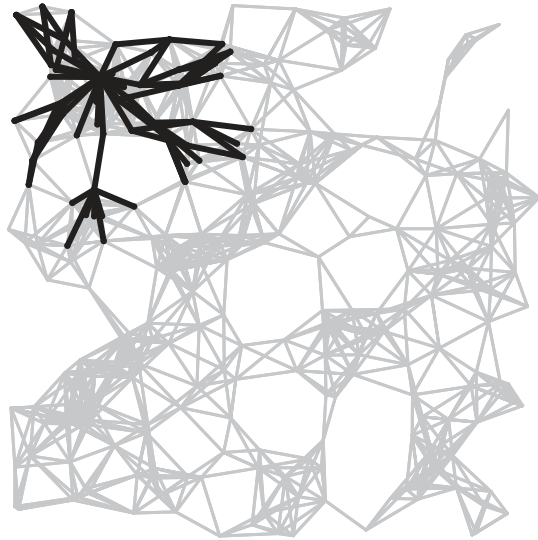


v	distTo[]	edgeTo[]
0	0.0	-
1	5.0	0→1
2	14.0	5→2
3	17.0	2→3
4	9.0	0→4
5	13.0	4→5
6	25.0	2→6
7	8.0	0→7

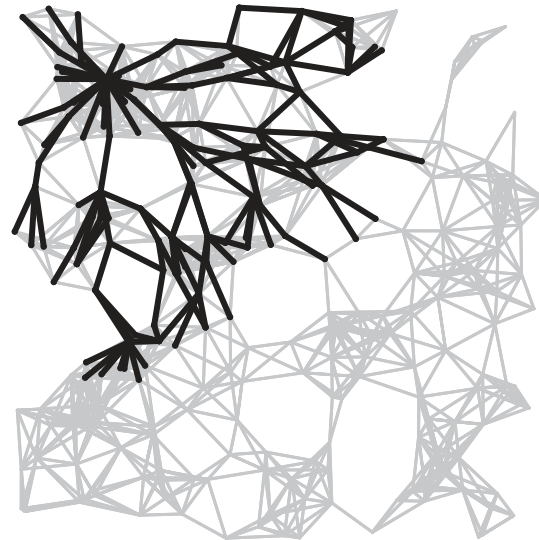
shortest-paths tree from vertex s

Bellman-Ford algorithm: visualization

passes
4



7



10



13



SPT



Bellman-Ford algorithm: analysis

Bellman-Ford algorithm

Initialize $\text{distTo}[s] = 0$ and $\text{distTo}[v] = \infty$ for all other vertices.

Repeat V times:

- Relax each edge.**
-

Proposition. Dynamic programming algorithm computes SPT in any edge-weighted digraph with no negative cycles in time proportional to $E \times V$.

Pf idea. After pass i , found path that is at least as short as any shortest path containing i (or fewer) edges.

Bellman-Ford algorithm: practical improvement

Observation. If $\text{distTo}[v]$ does not change during pass i , no need to relax any edge pointing from v in pass $i+1$.

FIFO implementation. Maintain **queue** of vertices whose $\text{distTo}[]$ changed.



be careful to keep at most one copy
of each vertex on queue (why?)

Overall effect.

- The running time is still proportional to $E \times V$ in worst case.
- But much faster than that in practice.

Single source shortest-paths implementation: cost summary

algorithm	restriction	typical case	worst case	extra space
topological sort	no directed cycles	$E + V$	$E + V$	V
Dijkstra (binary heap)	no negative weights	$E \log V$	$E \log V$	V
Bellman–Ford	no negative cycles	$E V$	$E V$	V
Bellman–Ford (queue-based)		$E + V$	$E V$	V

Remark 1. Directed cycles make the problem harder.

Remark 2. Negative weights make the problem harder.

Remark 3. Negative cycles makes the problem intractable.

Finding a negative cycle

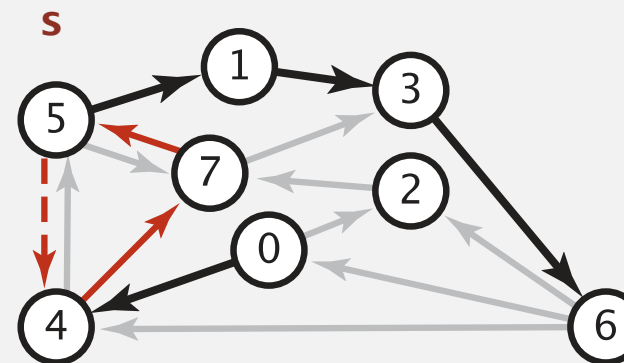
Negative cycle. Add two methods to the API for SP.

boolean hasNegativeCycle() *is there a negative cycle?*

Iterable <DirectedEdge> negativeCycle() *negative cycle reachable from s*

digraph

4->5 0.35
5->4 -0.66
4->7 0.37
5->7 0.28
7->5 0.28
5->1 0.32
0->4 0.38
0->2 0.26
7->3 0.39
1->3 0.29
2->7 0.34
6->2 0.40
3->6 0.52
6->0 0.58
6->4 0.93

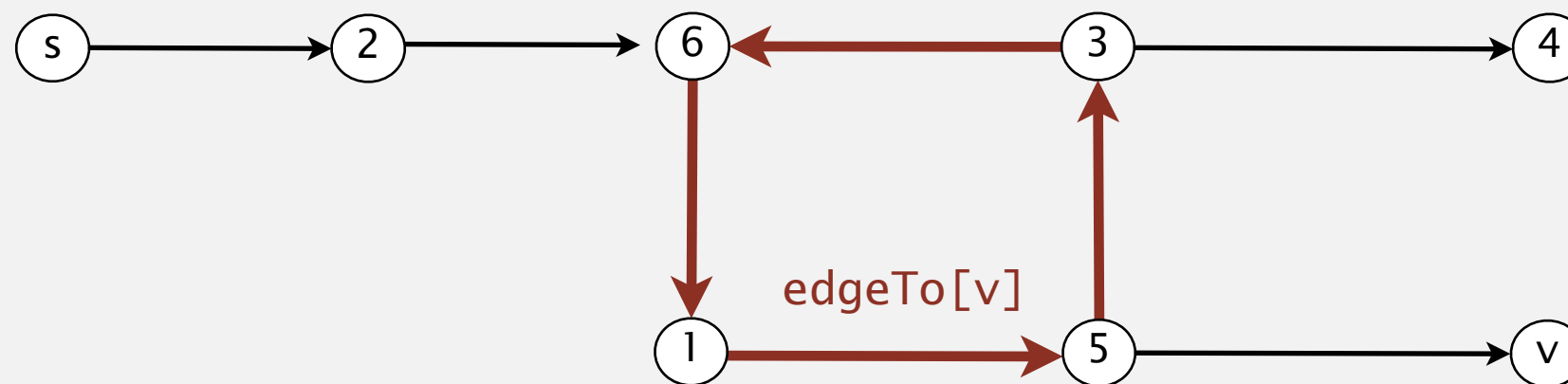


negative cycle (-0.66 + 0.37 + 0.28)

5->4->7->5

Finding a negative cycle

Observation. If there is a negative cycle, Bellman-Ford gets stuck in loop, updating `distTo[]` and `edgeTo[]` entries of vertices in the cycle.



Proposition. If any vertex v is updated in pass v , there exists a negative cycle (and can trace back `edgeTo[v]` entries to find it).

In practice. Check for negative cycles more frequently.

Negative cycle application: arbitrage detection

Problem. Given table of exchange rates, is there an arbitrage opportunity?

	USD	EUR	GBP	CHF	CAD
USD	1	0.741	0.657	1.061	1.011
EUR	1.350	1	0.888	1.433	1.366
GBP	1.521	1.126	1	1.614	1.538
CHF	0.943	0.698	0.620	1	0.953
CAD	0.995	0.732	0.650	1.049	1

Ex. \$1,000 $\Rightarrow$ 741 Euros $\Rightarrow$ 1,012.206 Canadian dollars $\Rightarrow$ \$1,007.14497.

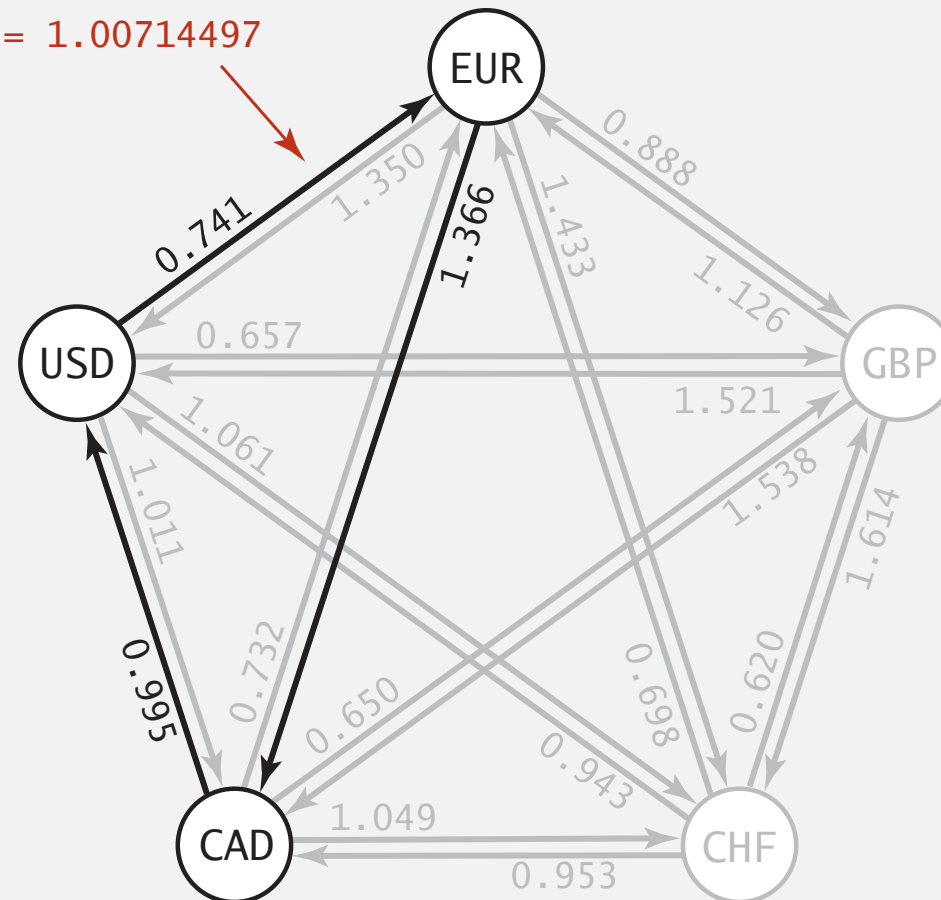

 $1000 \times 0.741 \times 1.366 \times 0.995 = 1007.14497$

Negative cycle application: arbitrage detection

Currency exchange graph.

- Vertex = currency.
- Edge = transaction, with weight equal to exchange rate.
- Find a directed cycle whose product of edge weights is > 1 .

$$0.741 * 1.366 * .995 = 1.00714497$$

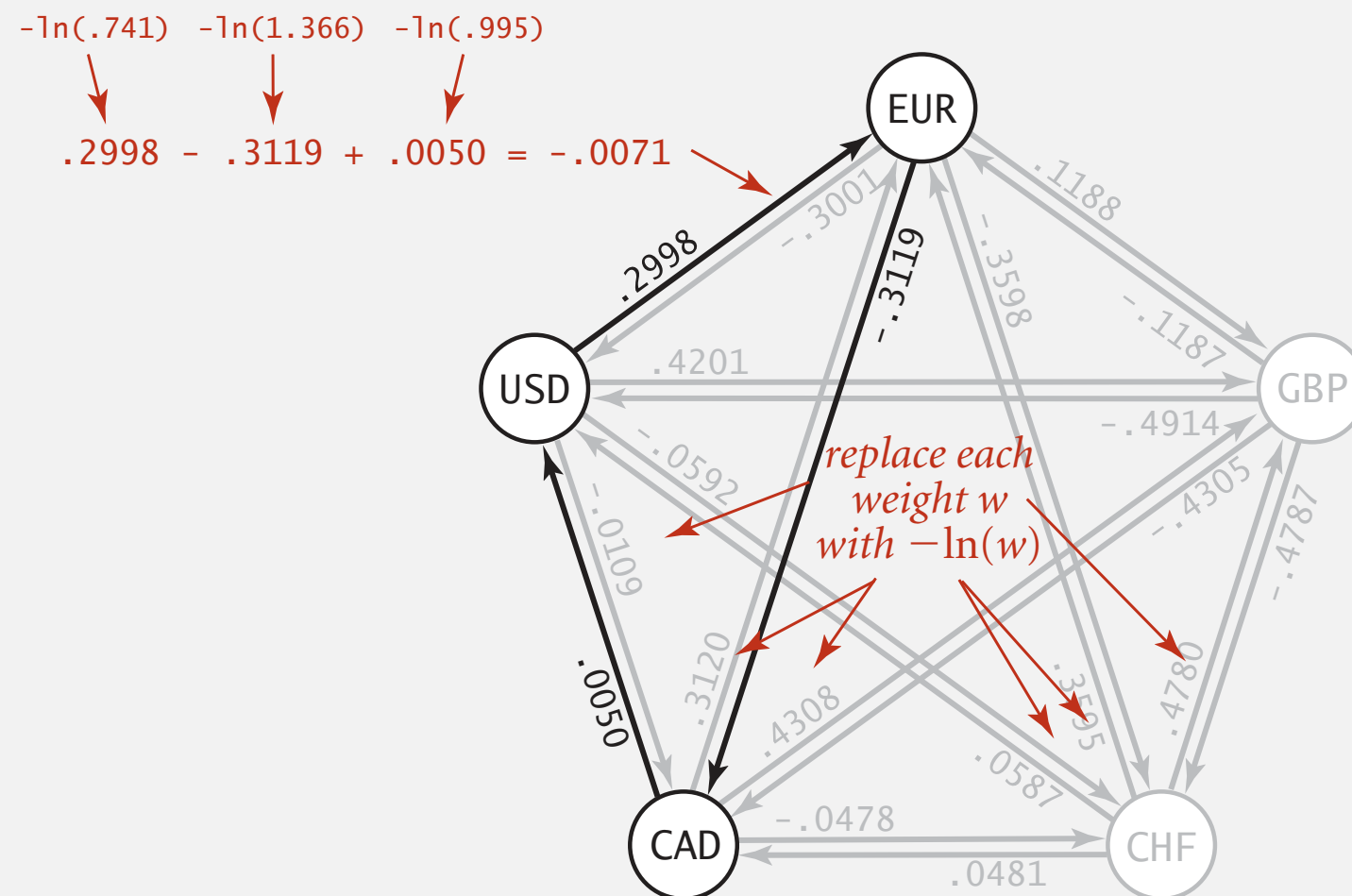


Challenge. Express as a negative cycle detection problem.

Negative cycle application: arbitrage detection

Model as a negative cycle detection problem by taking logs.

- Let weight of edge $v \rightarrow w$ be $-\ln$ (exchange rate from currency v to w).
- Multiplication turns to addition; > 1 turns to < 0 .
- Find a directed cycle whose sum of edge weights is < 0 (negative cycle).



Remark. Fastest algorithm is extraordinarily valuable!

Shortest paths summary

Nonnegative weights.

- Arises in many applications.
- Dijkstra's algorithm is nearly linear-time.

Acyclic edge-weighted digraphs.

- Arise in some applications.
- Topological sort algorithm is linear time.
- Edge weights can be negative.

Negative weights and negative cycles.

- Arise in some applications.
- If no negative cycles, can find shortest paths via Bellman-Ford.
- If negative cycles, can find one via Bellman-Ford.

Shortest-paths is a broadly useful problem-solving model.